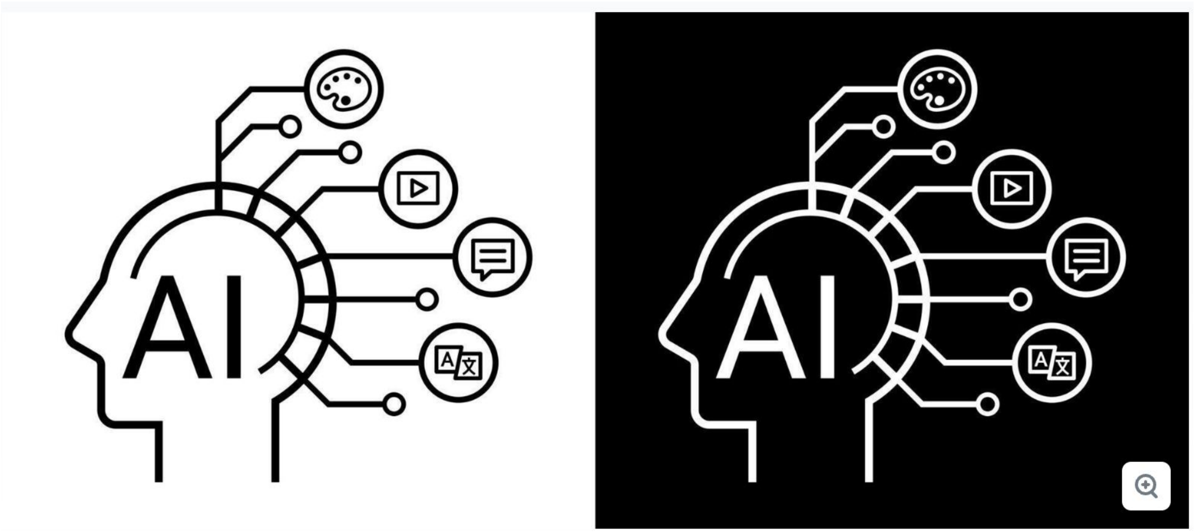
AIMS 5740
Generative Artificial Intelligence

(2026 Term 2)

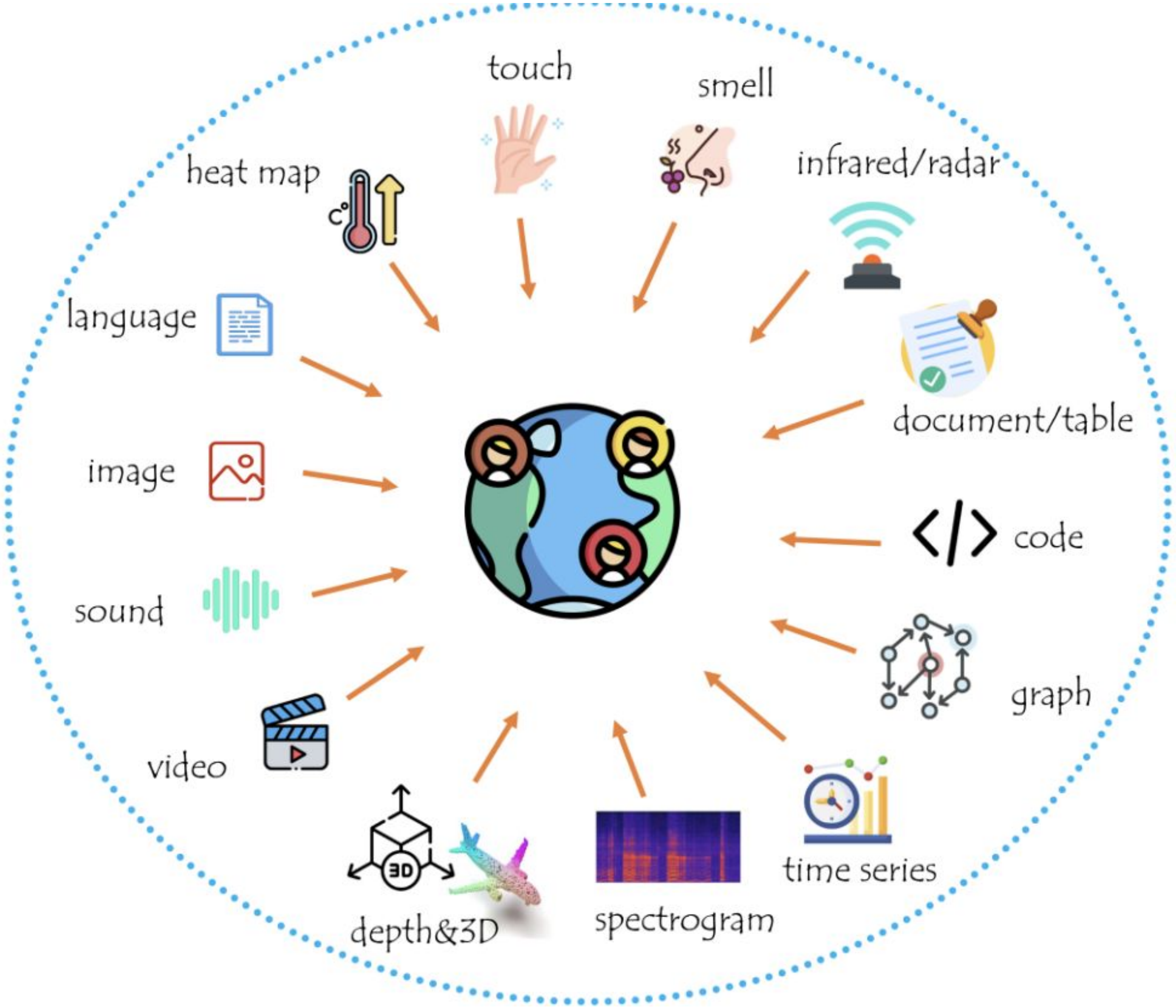


Computer Science & Engineering
The Chinese University of Hong Kong

Intelligence in Multi-Sensory Data

Harnessing Multimodality

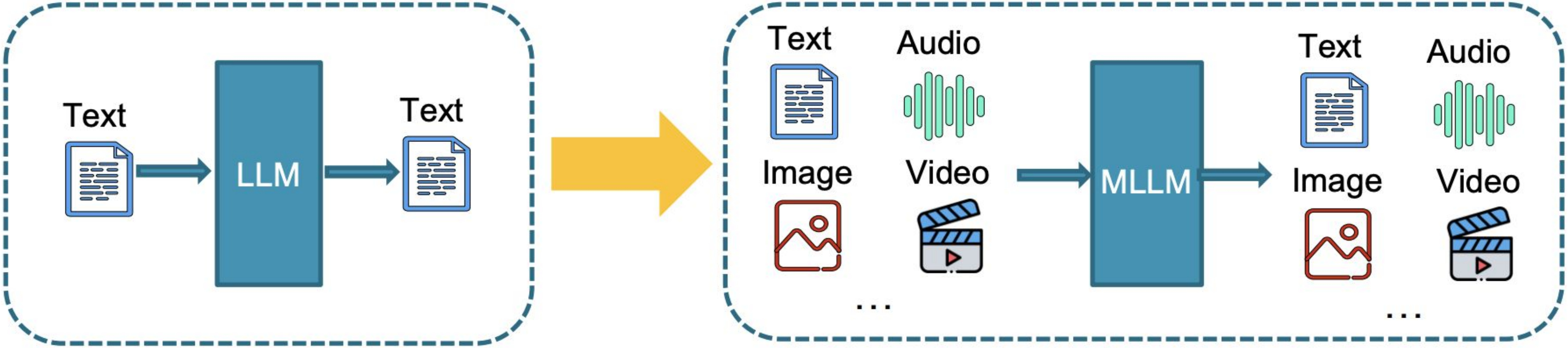
☞ This world we live in is replete with multimodal information & signals, not just language.



Intelligence in Multi-Sensory Data

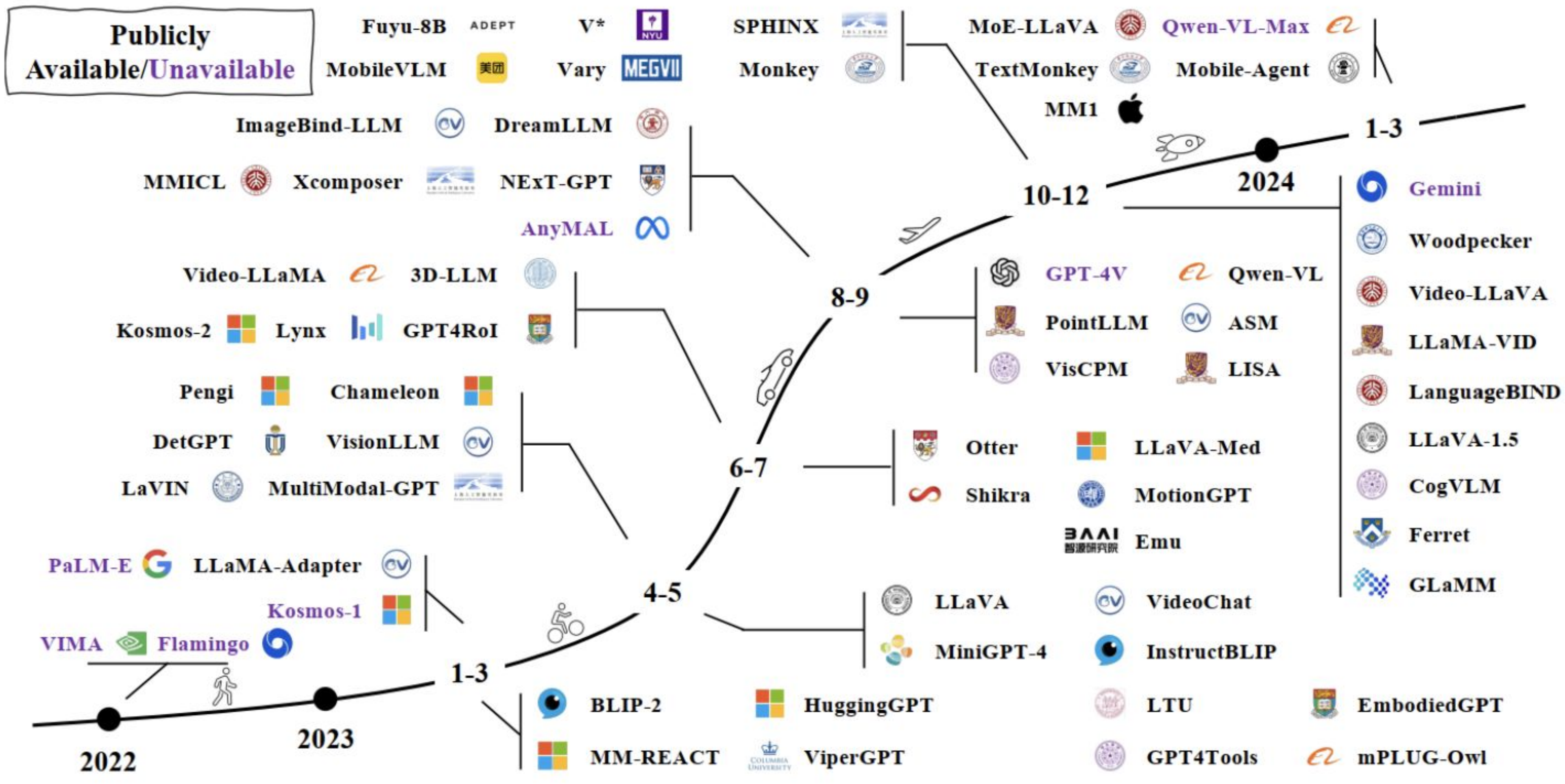
Building Multimodal LLMs (MLLMs)

Can we transfer the success of LLMs to MLLMs, enabling LLMs to comprehend multimodal information as deeply as they understand language?



Intelligence in Multi-Sensory Data

Trends of MLLMs



LLaVA

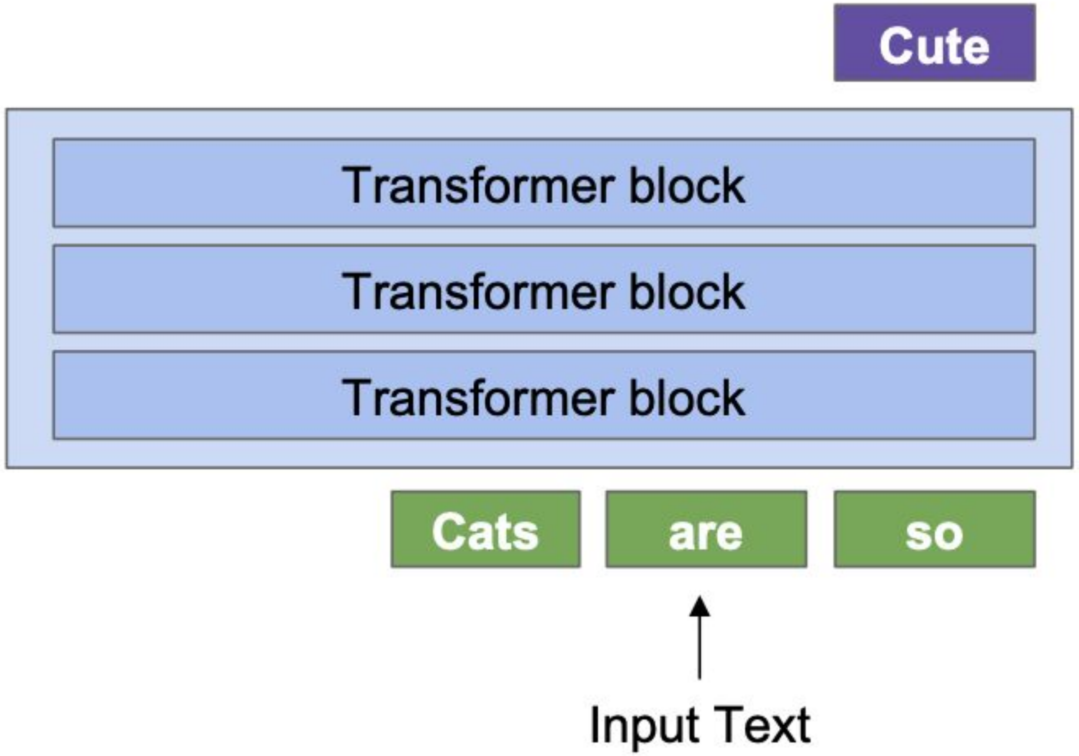
Motivation: Language models which do next token prediction can be applied to a wide variety of tasks at inference (Math, sentiment analysis, symbolic reasoning)

Can we build a model that can accept images and text as input, and then output text?

→ Vision-Language Models

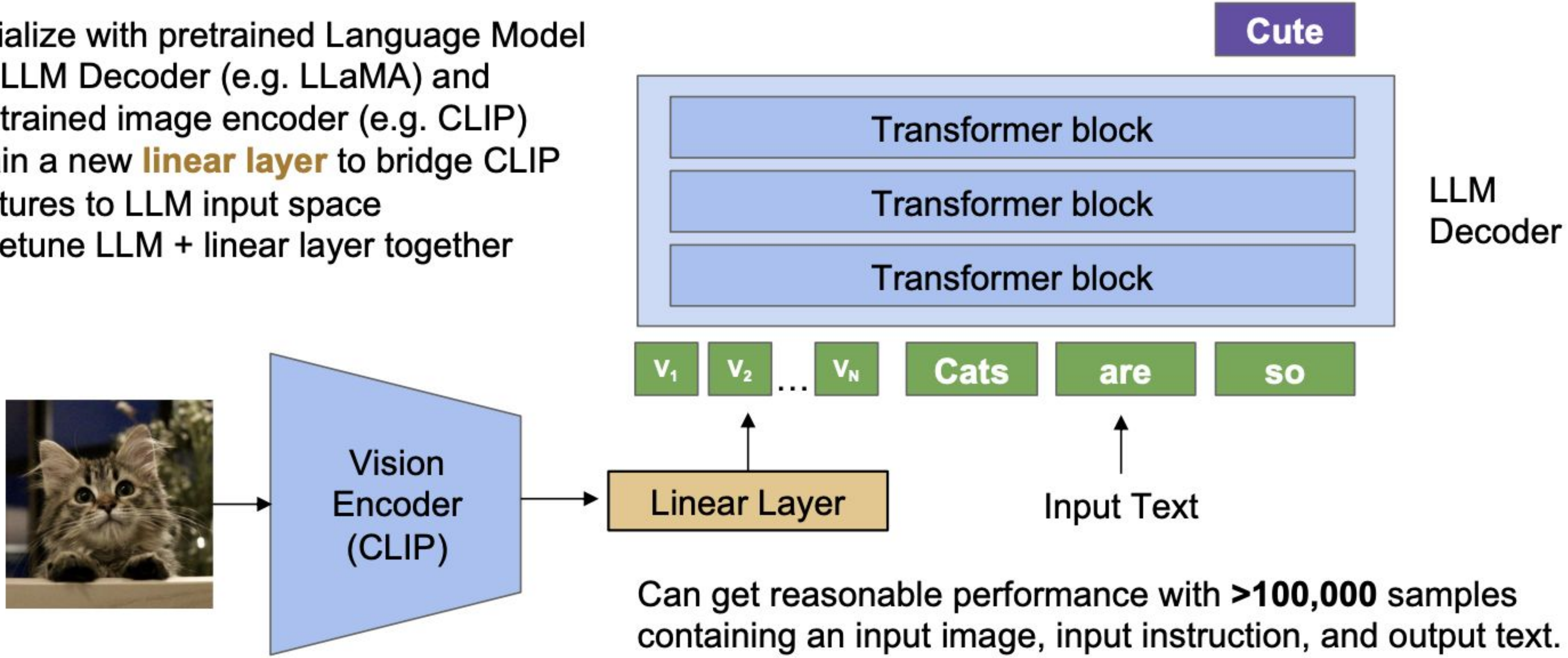
LLaVA

Recall how transformers decode language



LLaVA – Overall Architecture + Training Recipe

1. Initialize with pretrained Language Model for LLM Decoder (e.g. LLaMA) and pretrained image encoder (e.g. CLIP)
2. Train a new **linear layer** to bridge CLIP features to LLM input space
3. Finetune LLM + linear layer together

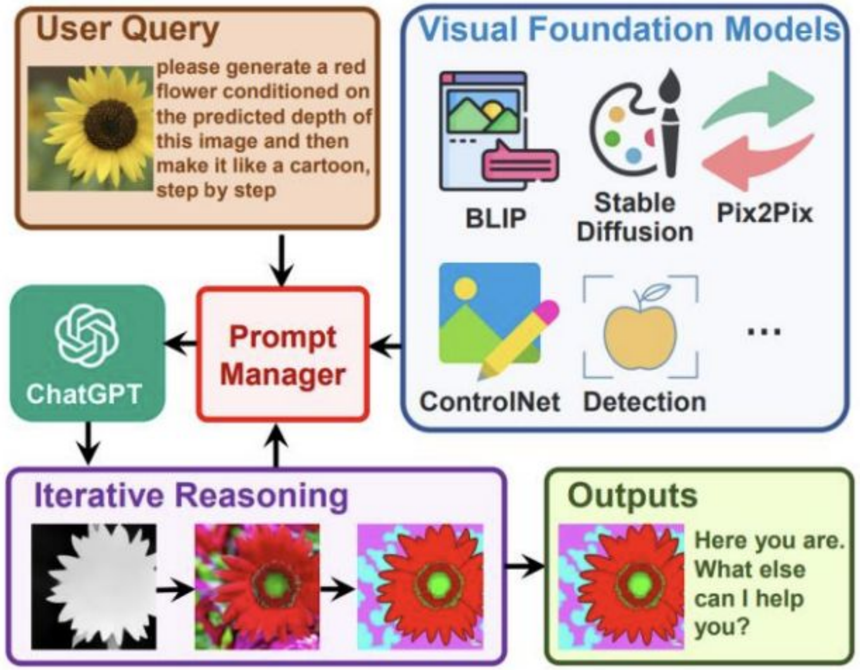


Can get reasonable performance with **>100,000** samples containing an input image, input instruction, and output text.

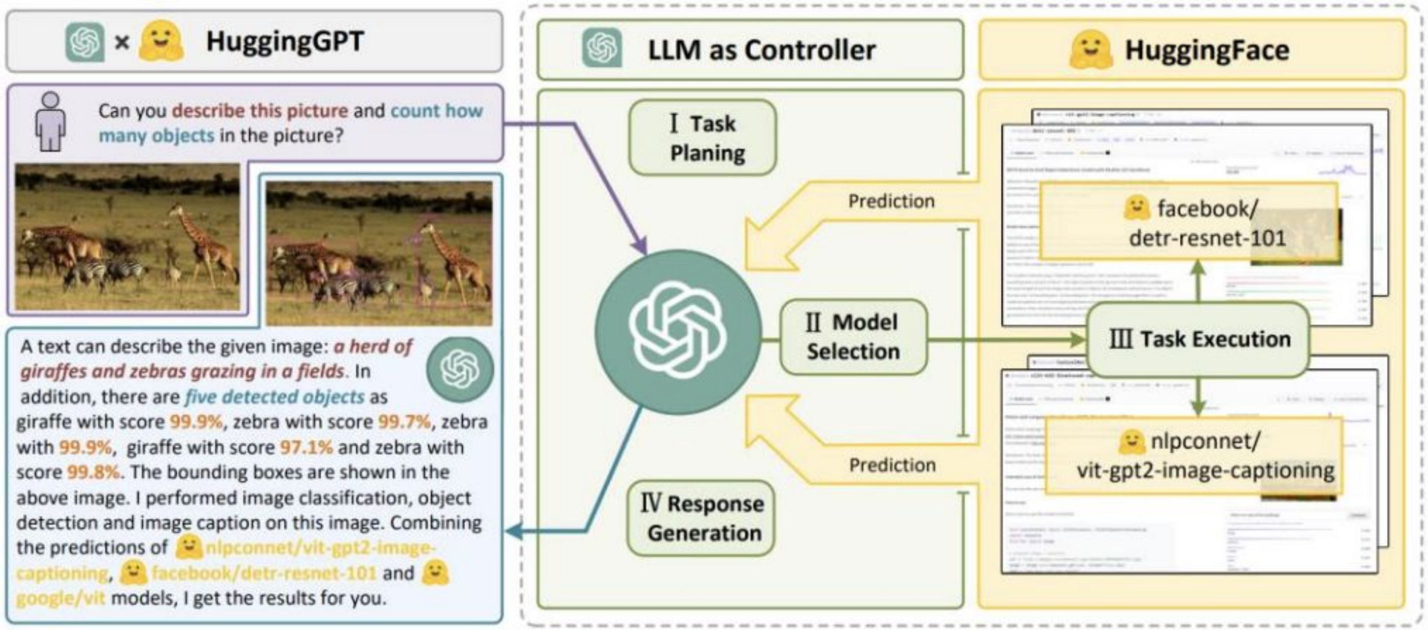
LLaVA – Other Forms

LLM as Discrete Scheduler/Controller

- Visual-ChatGPT



- HuggingGPT



- Quick to build (without training), flexible extension to many tool features
- Information loss in text medium, the bottle-neck

LLaVA – Encoding

- High-resolution Multimodal LLMs
  - Image slice-based: Split high-resolution images into slices
  - Representatives:
    - GPT-4V, LLaVA-NeXT, MiniCPM-V 2.0/2.5, LLaVA-UHD, mPLUG-DocOwl 1.5, SPHINX, InternLM-XComposer2-4KHD, Monkey

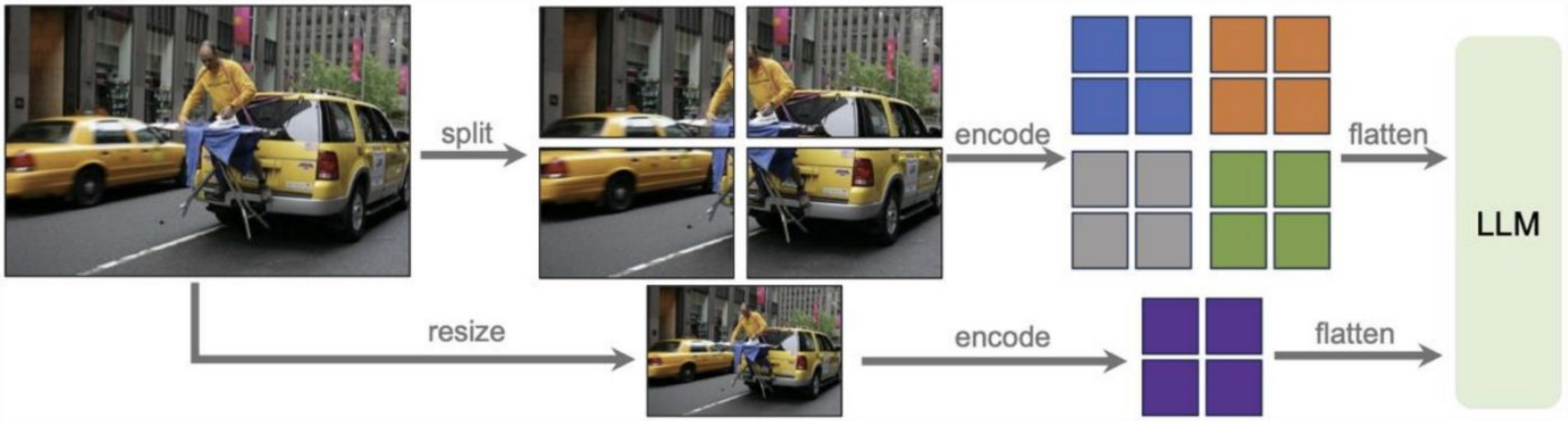


Illustration of dynamic high resolution scheme: a grid configuration of 2×2

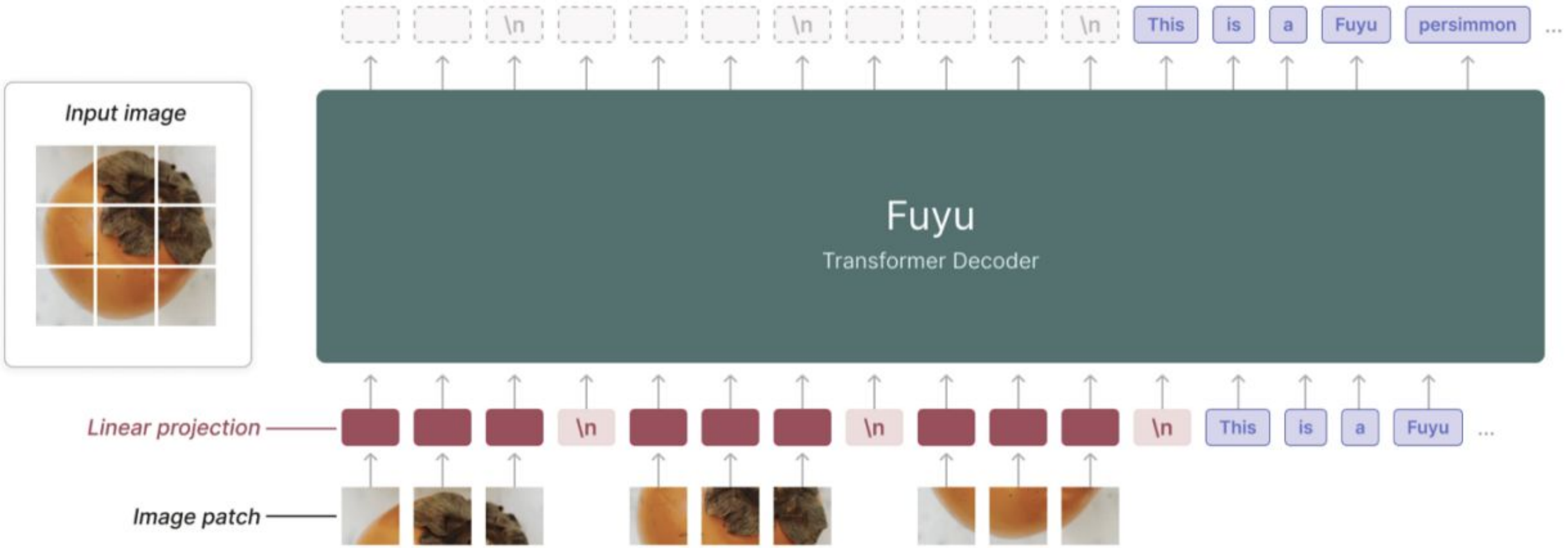
LLaVA – Encoding

- High-resolution Multimodal LLMs
  - Image slice-based: Split high-resolution images into slices
  - OCR capabilities improves significantly without new data

Model	#Data	MaxRes.	AR.	TFLOPs	VQA^{v2}	GQA	VQA^{T}	POPE	SQA	VizWiz	MME	MMB	MMB^{CN}
BLIP-2 [21]	129M	224×224	Fix	1.0	41.0	41.0	42.5	85.3	61.0	19.6	1293.8	-	-
InstructBLIP [11]	130M	224×224	Fix	1.0	-	49.5	50.7	78.9	63.1	33.4	1212.8	-	-
Shikra [8]	6M	224×224	Fix	8.0	77.4	-	-	-	-	-	-	58.8	-
Qwen-VL [5]	1.4B	448×448	Fix	9.2	78.8	59.3	<u>63.8</u>	-	67.1	35.2	-	38.2	7.4
SPHINX [24]	1.0B	448×448	Fix	39.7	78.1	62.6	51.6	80.7	69.3	39.9	1476.1	66.9	56.2
SPHINX-2k [24]	1.0B	762×762	Fix	69.4	<u>80.7</u>	63.1	61.2	<u>87.2</u>	70.6	44.9	1470.7	65.9	57.9
MiniGPT-v2 [7]	326M	448×448	Fix	4.3	-	60.1	-	-	-	53.6	-	-	-
Fuyu-8B [6]	-	1024×1024	Any	21.3	74.2	-	-	74.1	-	-	728.6	10.7	-
OtterHD-8B [20]	-	1024×1024	Any	21.3	-	-	-	86.0	-	-	1223.4	58.3	-
mPLUG-Owl2 [43]	401M	448×448	Fix	1.7	79.4	56.1	58.2	86.2	68.7	54.5	1450.2	64.5	-
UReader [42]	86M	896×1120	Enum	26.0	-	-	57.6	-	-	-	-	-	-
Monkey [23]	1.0B	896×1344	Enum	65.3	80.3	60.7	-	67.6	69.4	**61.2**	-	-	-
LLaVA-1.5 [27]	1.2M	336×336	Fix	15.5	80.0	<u>63.3</u>	61.3	85.9	<u>71.6</u>	53.6	<u>1531.3</u>	<u>67.7</u>	<u>63.6</u>
LLaVA-UHD (ours)	1.2M	672×1008	Any	14.6	**81.7**	**65.2**	**67.7**	**89.1**	**72.0**	<u>56.1</u>	**1535.0**	**68.0**	**64.8**
Δ	-	×6 times	-	-0.9	+1.7	+1.9	+6.4	+3.2	+0.4	+2.5	+3.7	+0.3	+1.2

LLaVA – Encoding

- High-resolution Multimodal LLMs
  - ViT-free: linear project pixel-patches into tokens
  - Representatives: **Fuyu, OtterHD**
  - A potential unified way for MLLMs, getting rid of ViTs
  - More costly to train, produce lengthy visual tokens

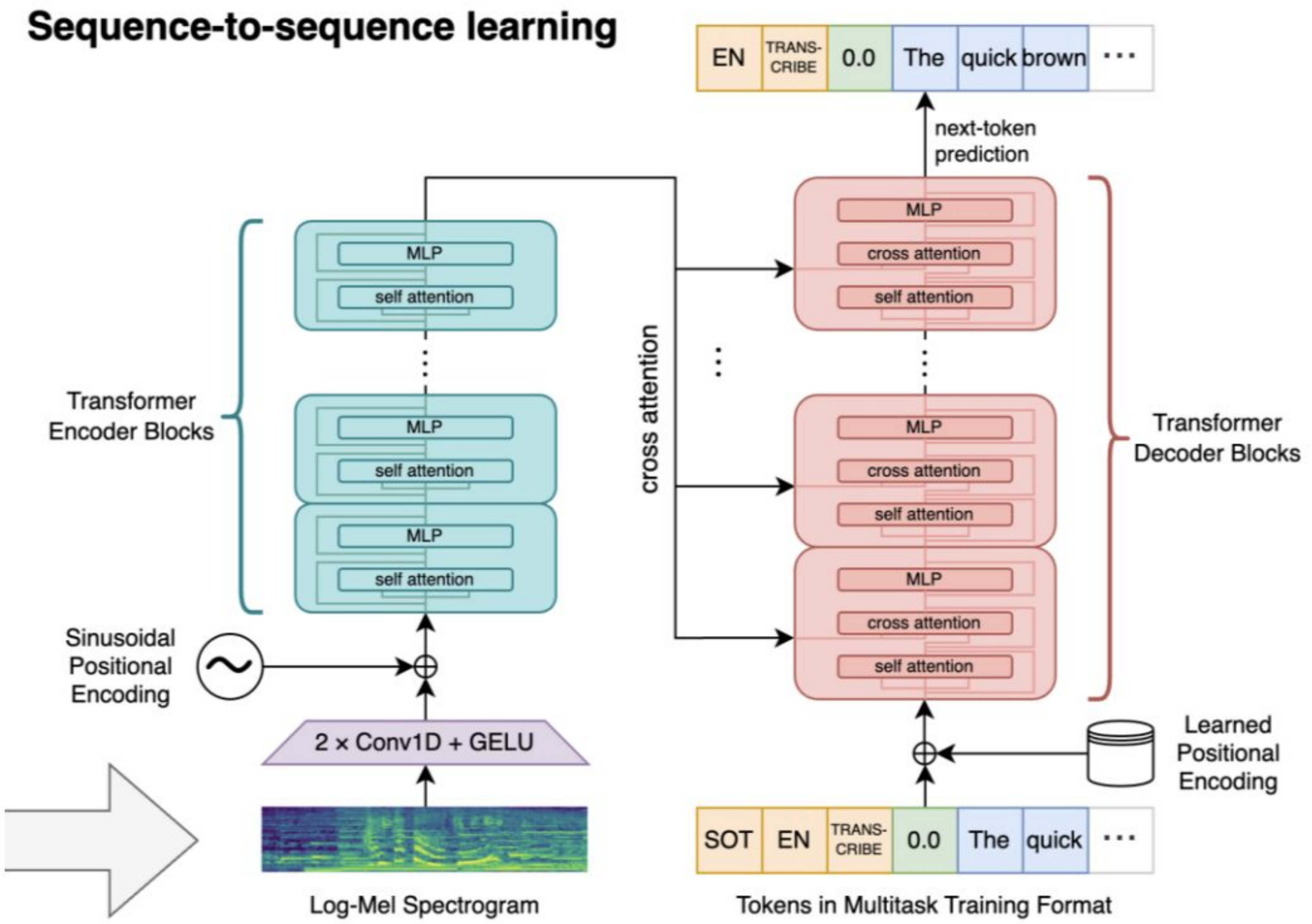


LLaVA – Multimodal Encoding

Non-Visual Encoder

- Audio:
  - Whisper
  - AudioCLIP
  - HuBERT
  - BEATs

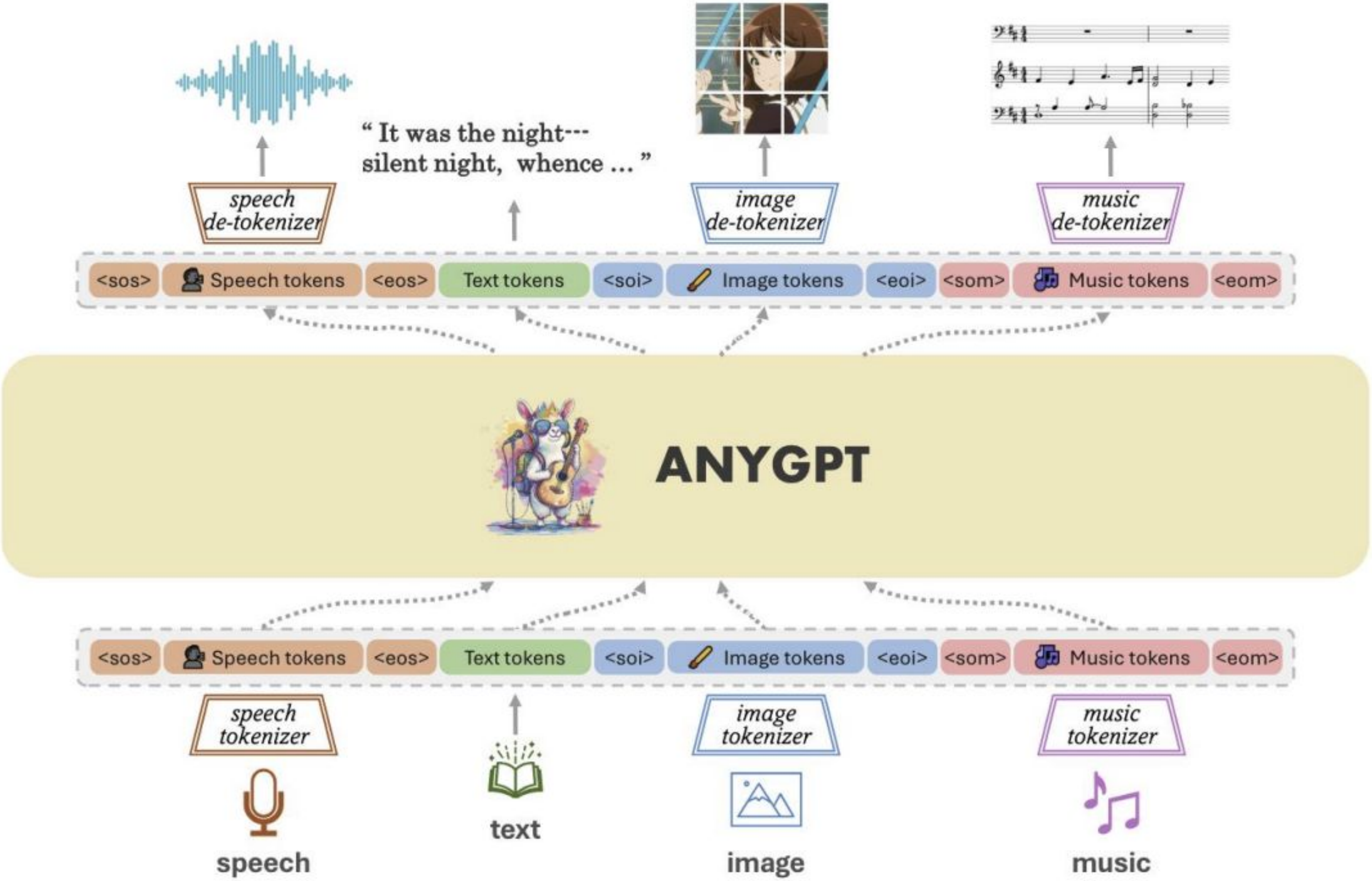
- 3D Point:
  - Point-BERT



LLaVA – Multimodal Encoding

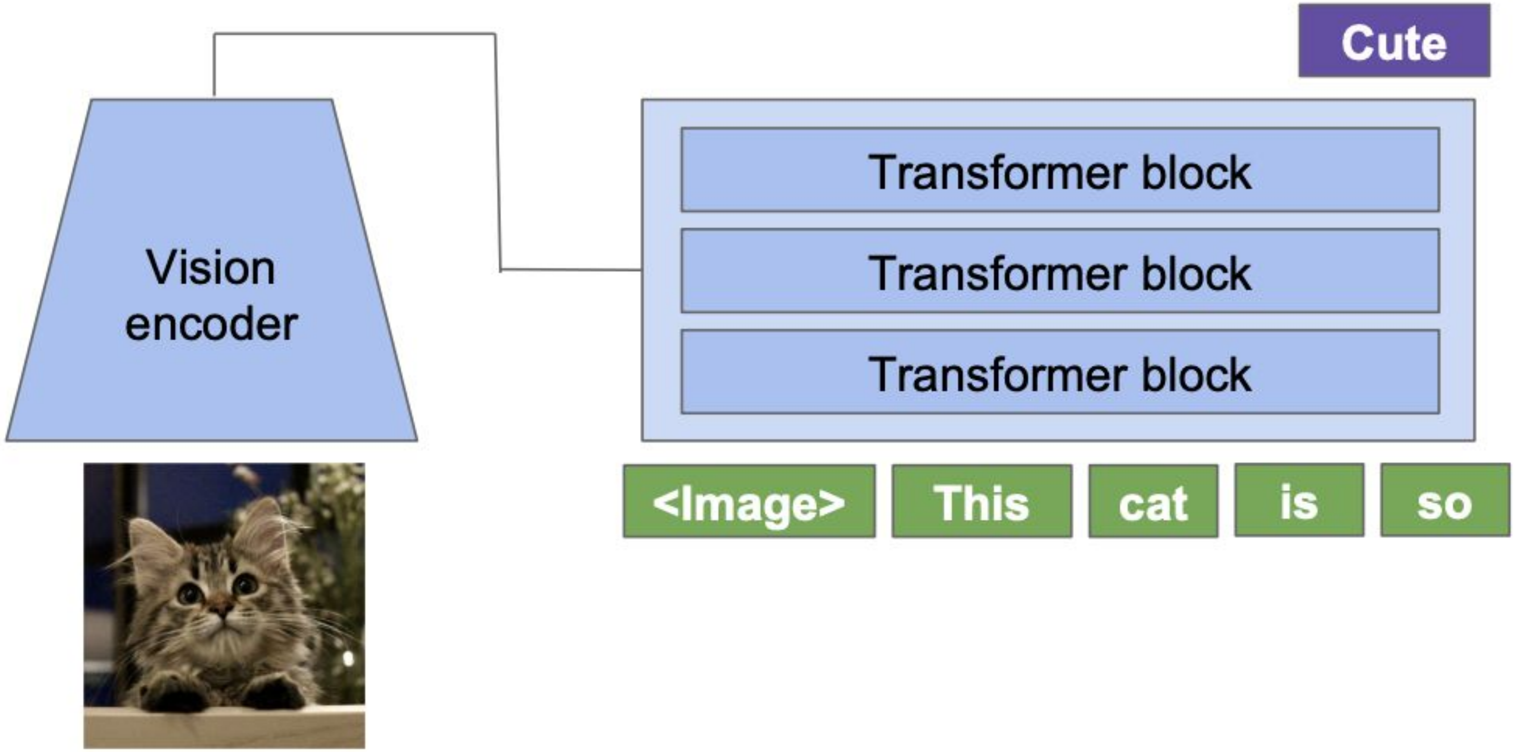
Tokenization

- AnyGPT

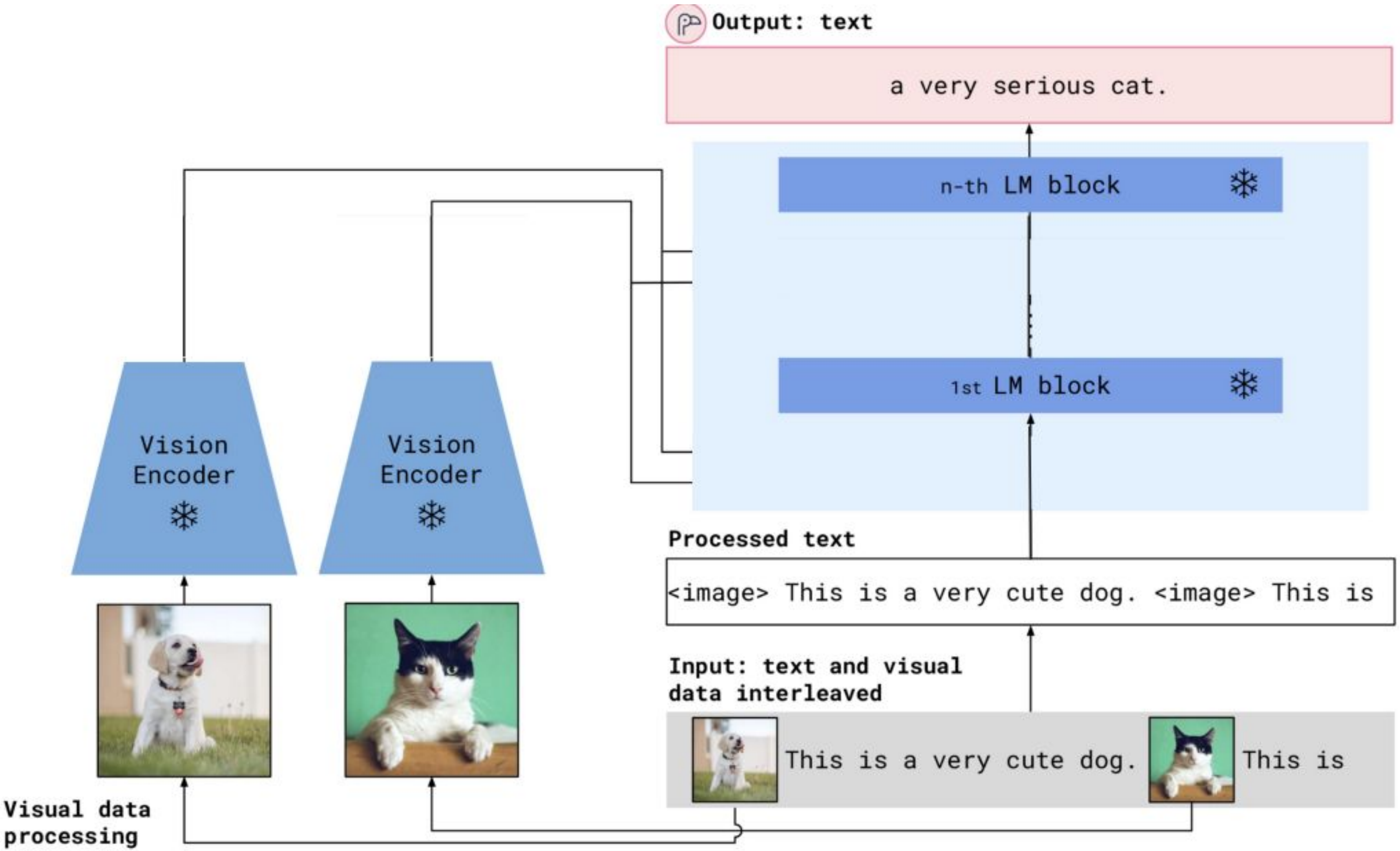


Flamingo

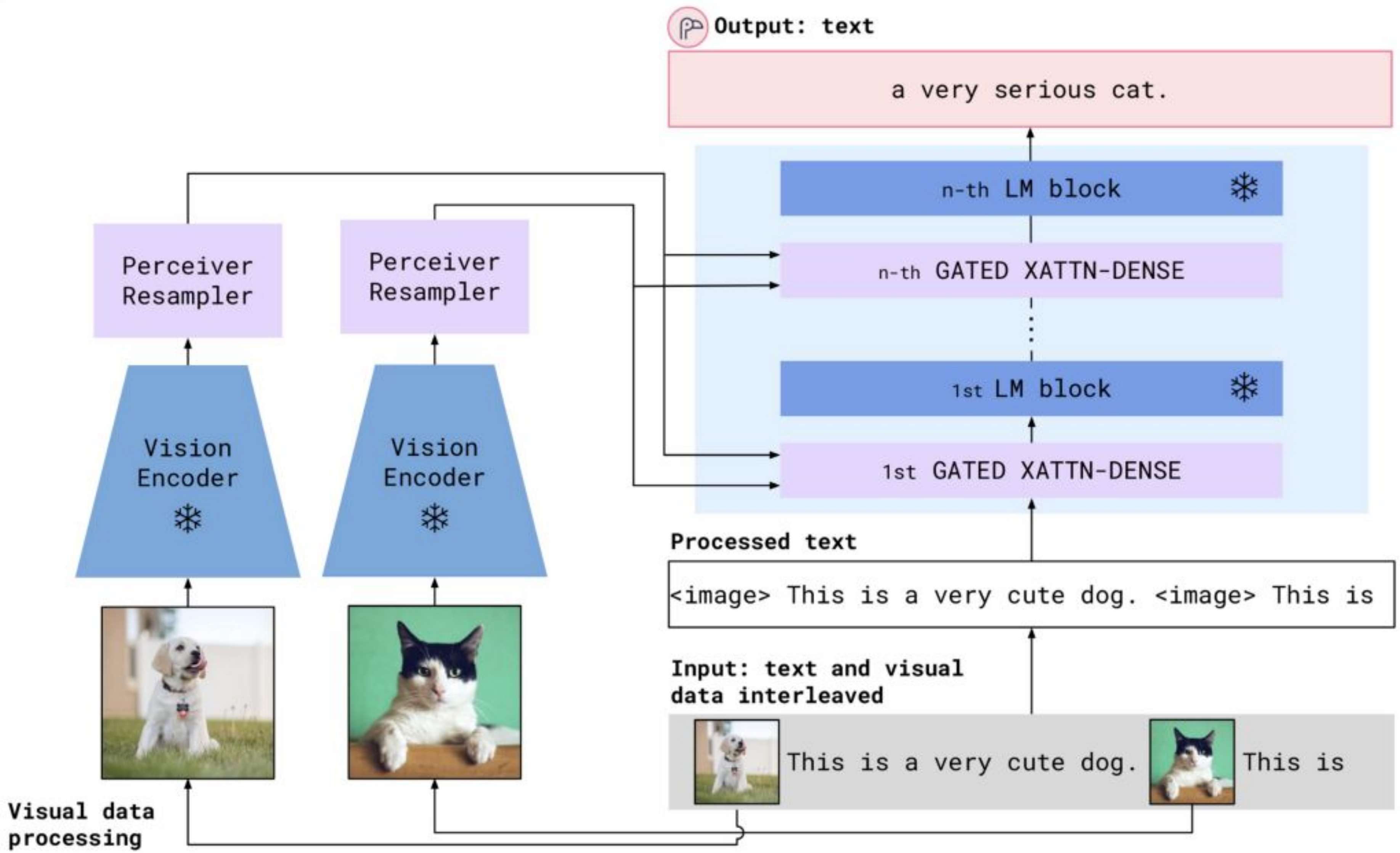
Flamingo followed up with a new way to fuse visual features



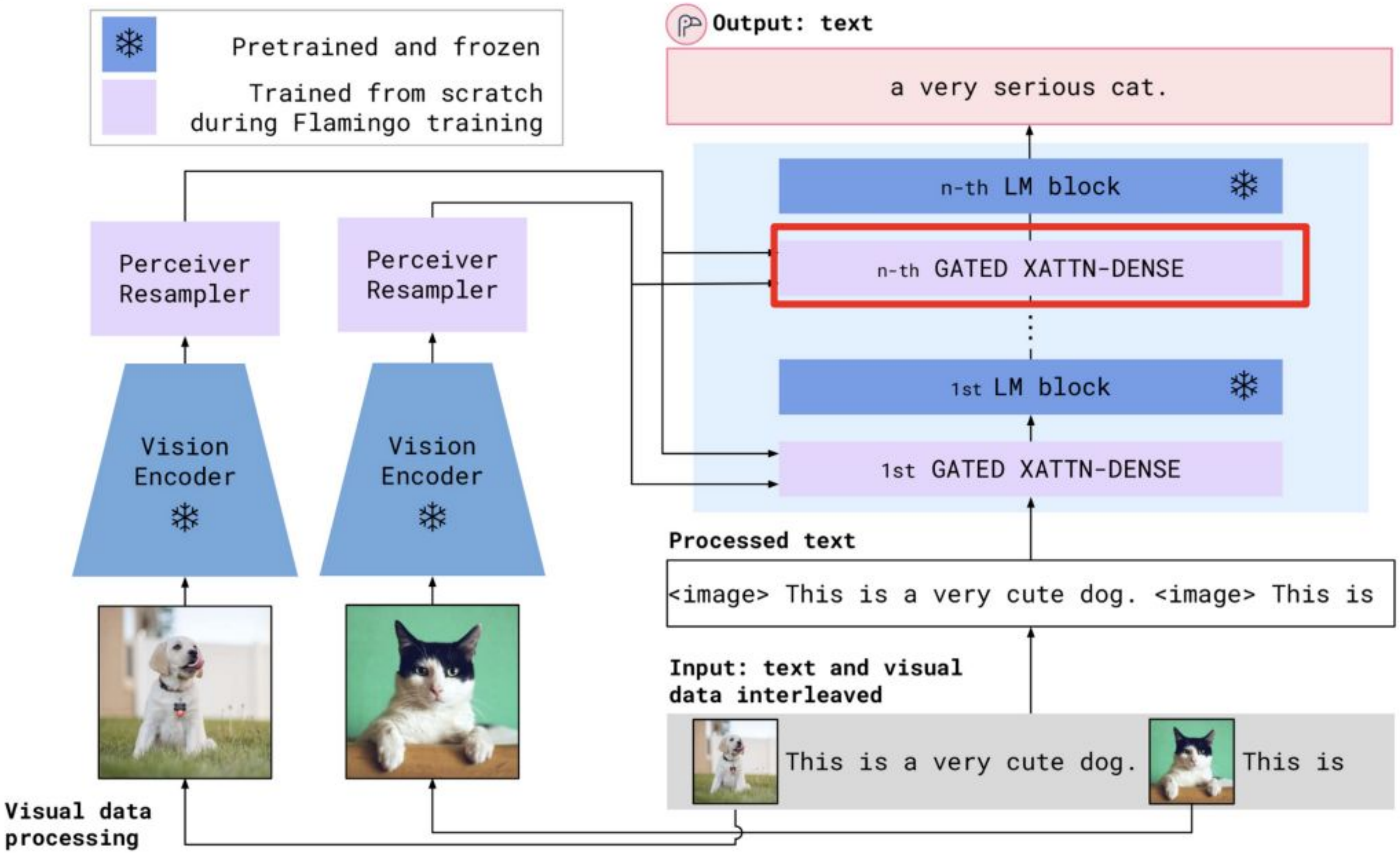
Pre-trained parts of Flamingo



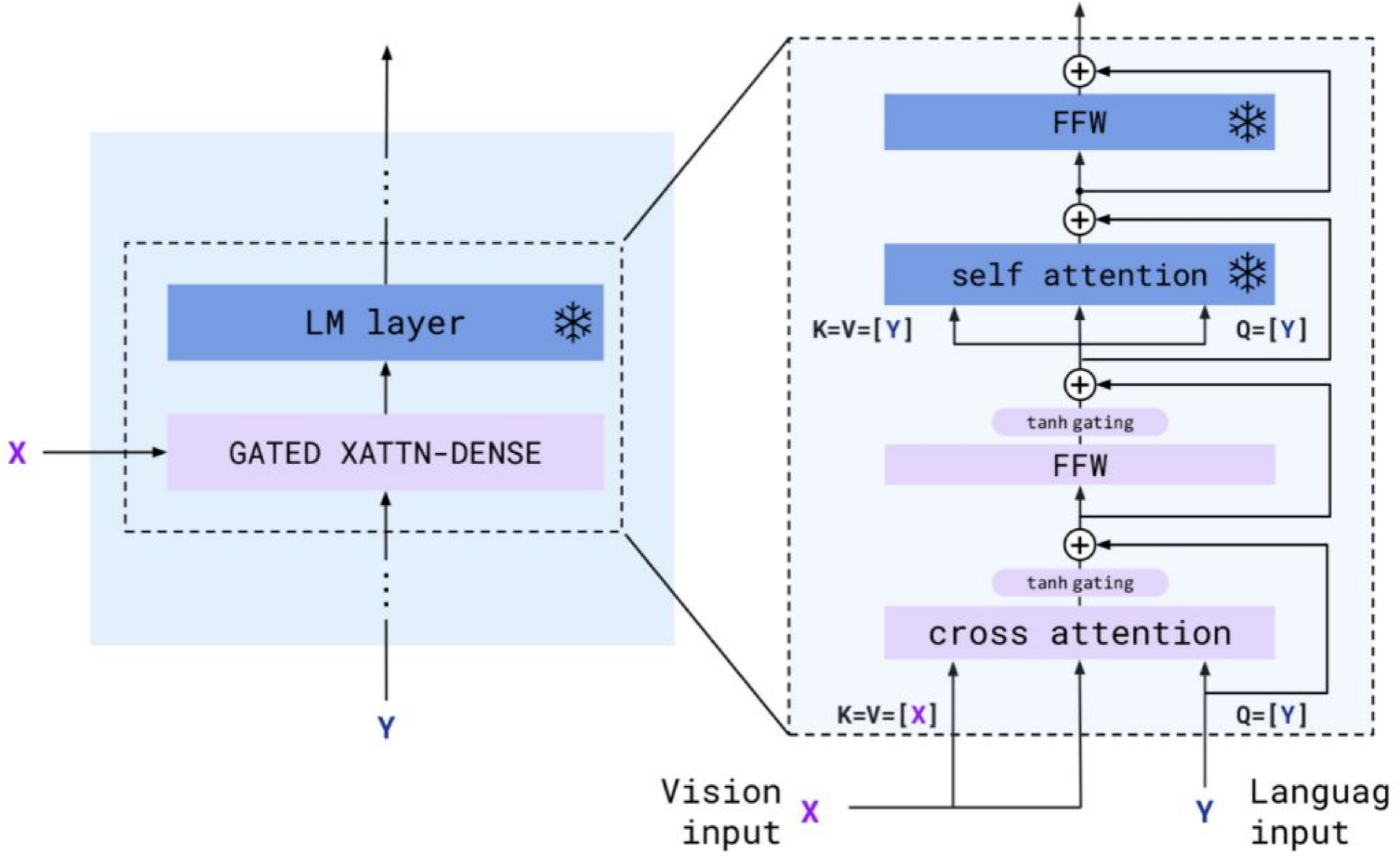
Trainable parts of Flamingo



Flamingo Full Architecture



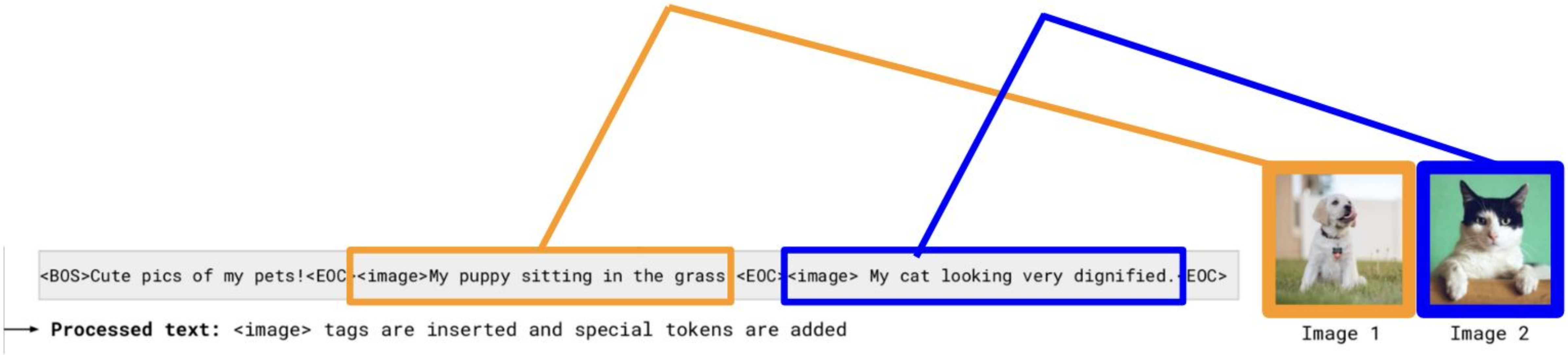
Gated Cross-Attention



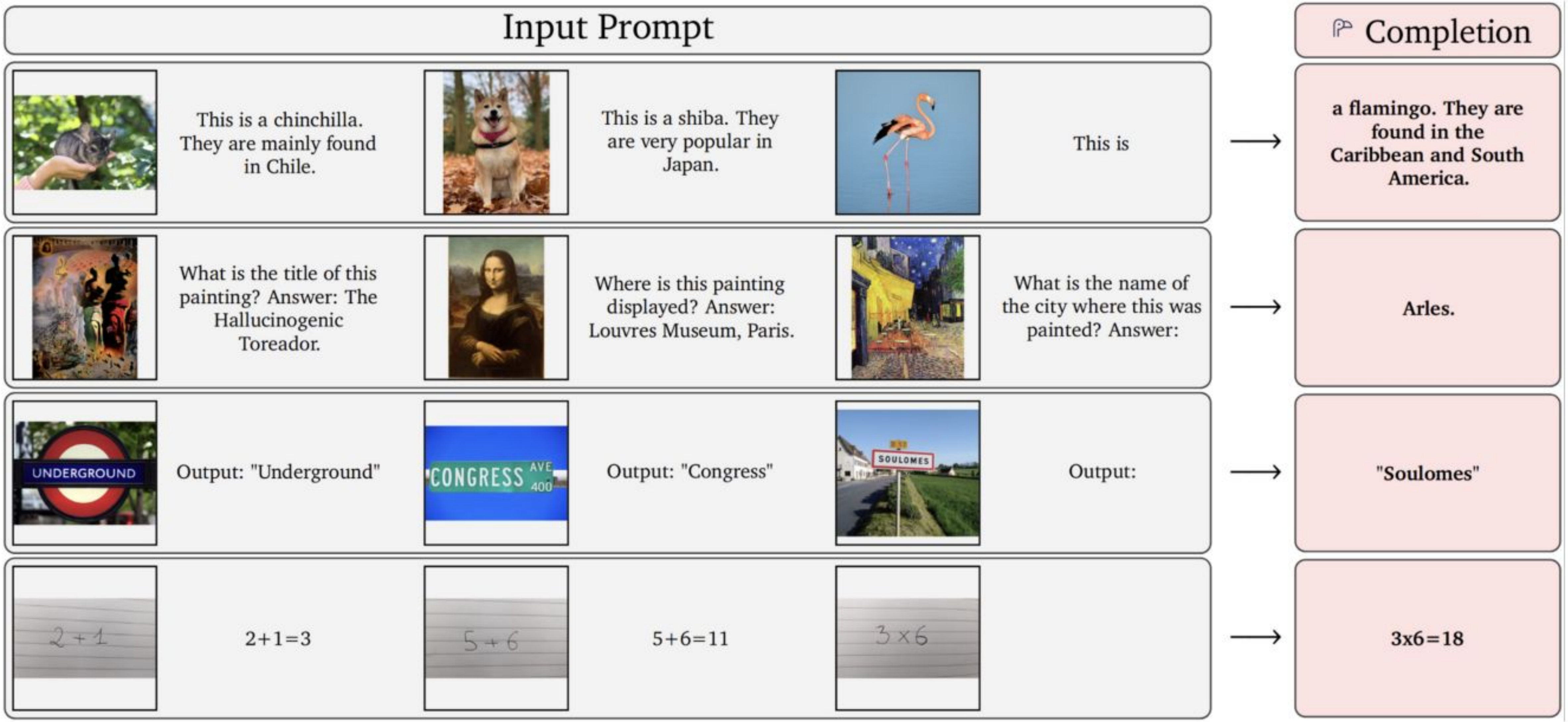
Flamingo: Training Data

Flamingo arranges its training data similar to language modeling,

with special tags <image>, <eos> to indicate when a new image shows up or the text ends.



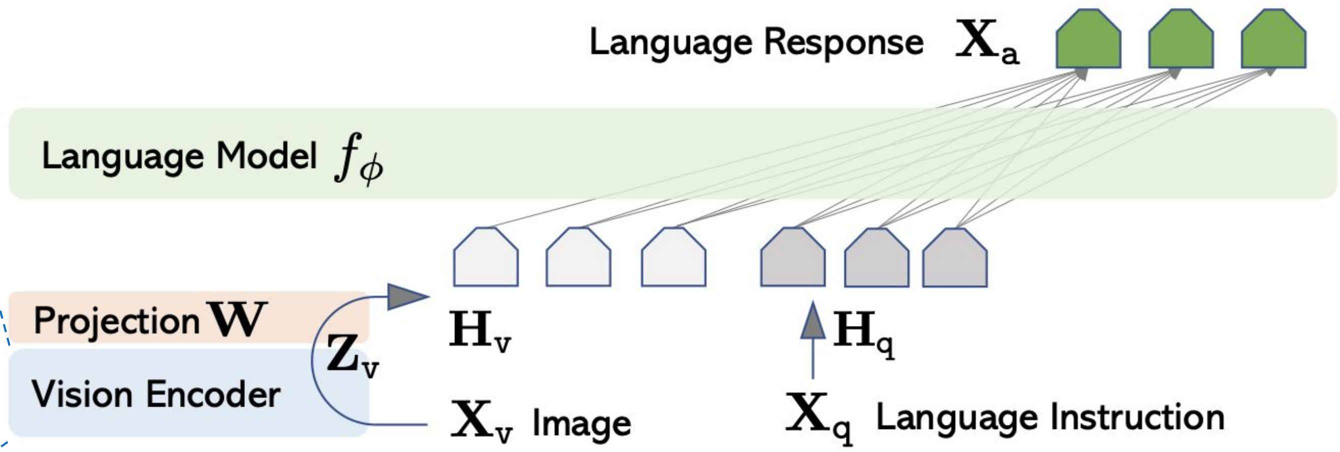
Flamingo: Results



CLIP as a Foundation Model in Multimodal Intelligence

Model	CLIP Used as the Vision Encoder
Flamingo[1]; Alayrac et al., Nov 2022	CLIP ViT-L/14
LLaVA[2]; Liu et al., Apr 2023	CLIP ViT-L/14@336px
Qwen-VL[3]; Bai et al., Oct 2023	Open-CLIP-bigG
InternVL[4]; Chen et al., Jan 2024	CLIP-6B
LLaVA-NeXT; Liu et al., Jan 2024	CLIP ViT-L/14@336px
Mini-Gemini[5]; Li et al., Mar 2024	CLIP ViT-L/14

LLaVA: A Typical Structure of MLLM



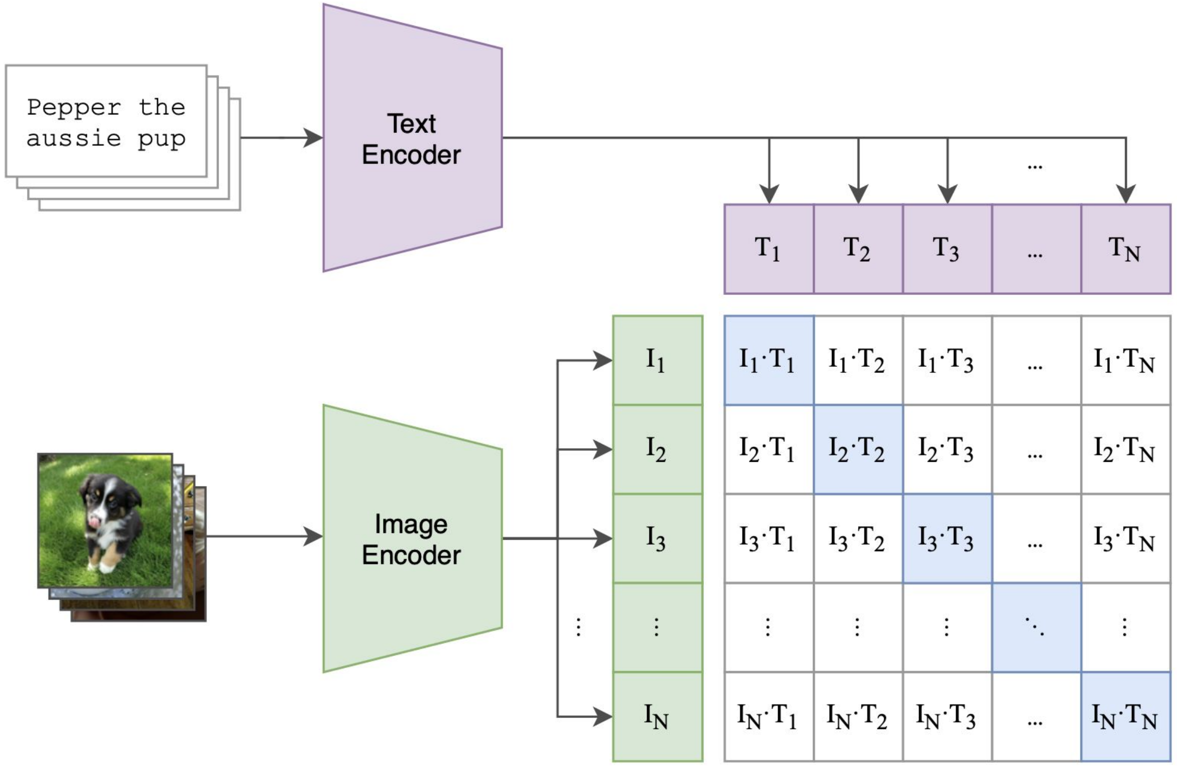
LLaVA; Liu et al., 2023

CLIP is broadly used as the vision encoder for various of MLLMs !

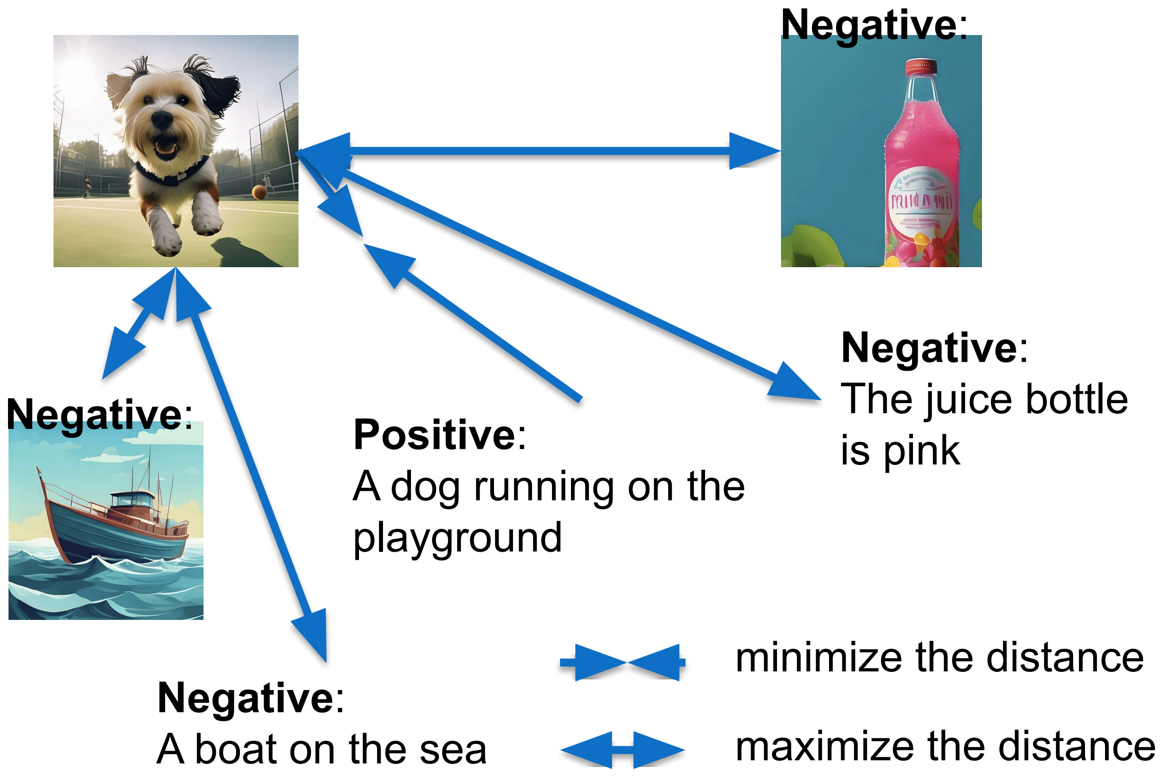
[1] *Flamingo: a visual language model for few-shot learning*
[2] *Visual Instruction Tuning*
[3] *A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond*
[4] *InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks*
[5] *LLaVA-NeXT: Improved reasoning, OCR, and world knowledge*
[6] *Mini-Gemini: Mining the Potential of Multi-modality Vision Language Models*

CLIP: Architecture and Training

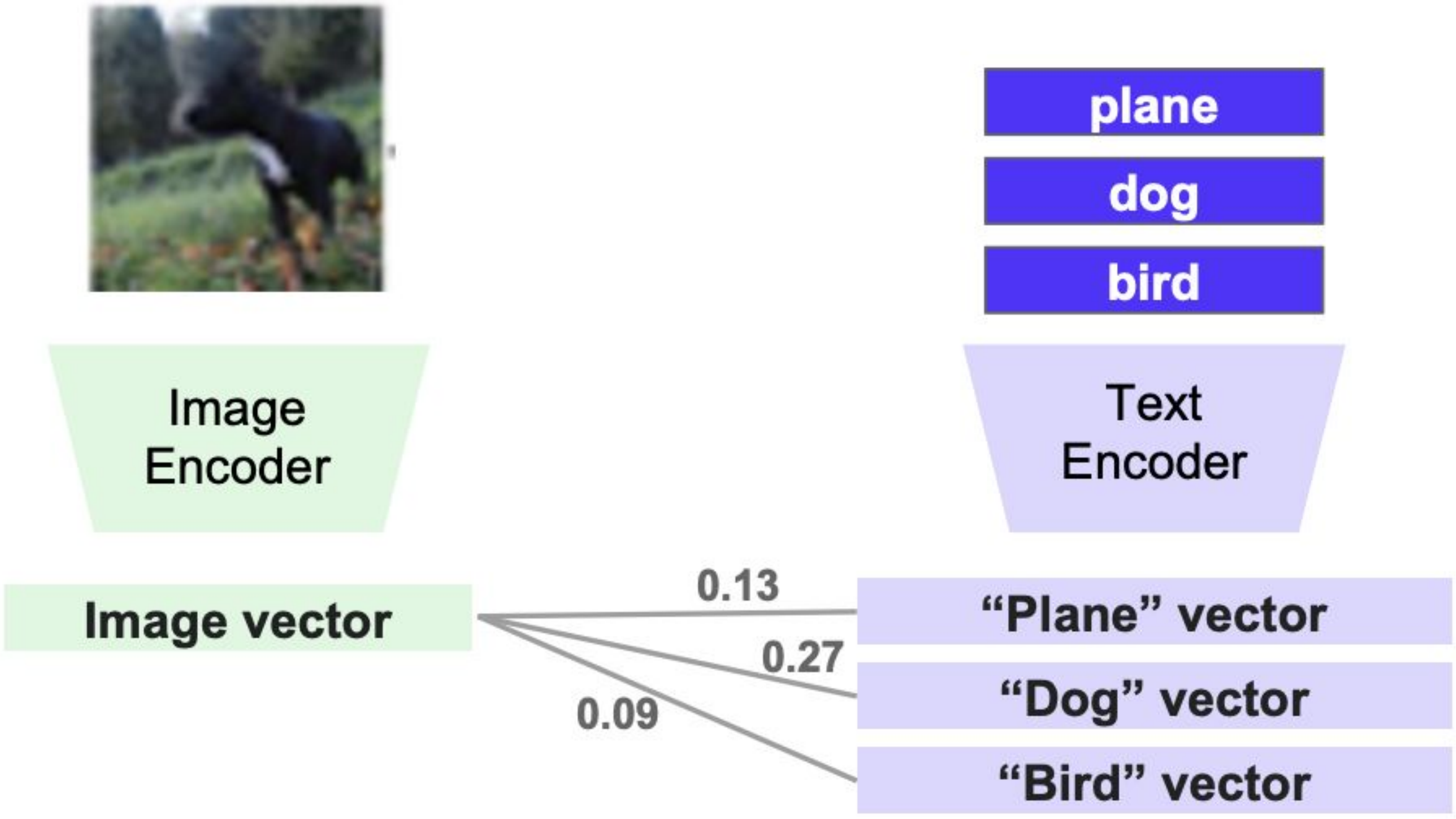
Architecture



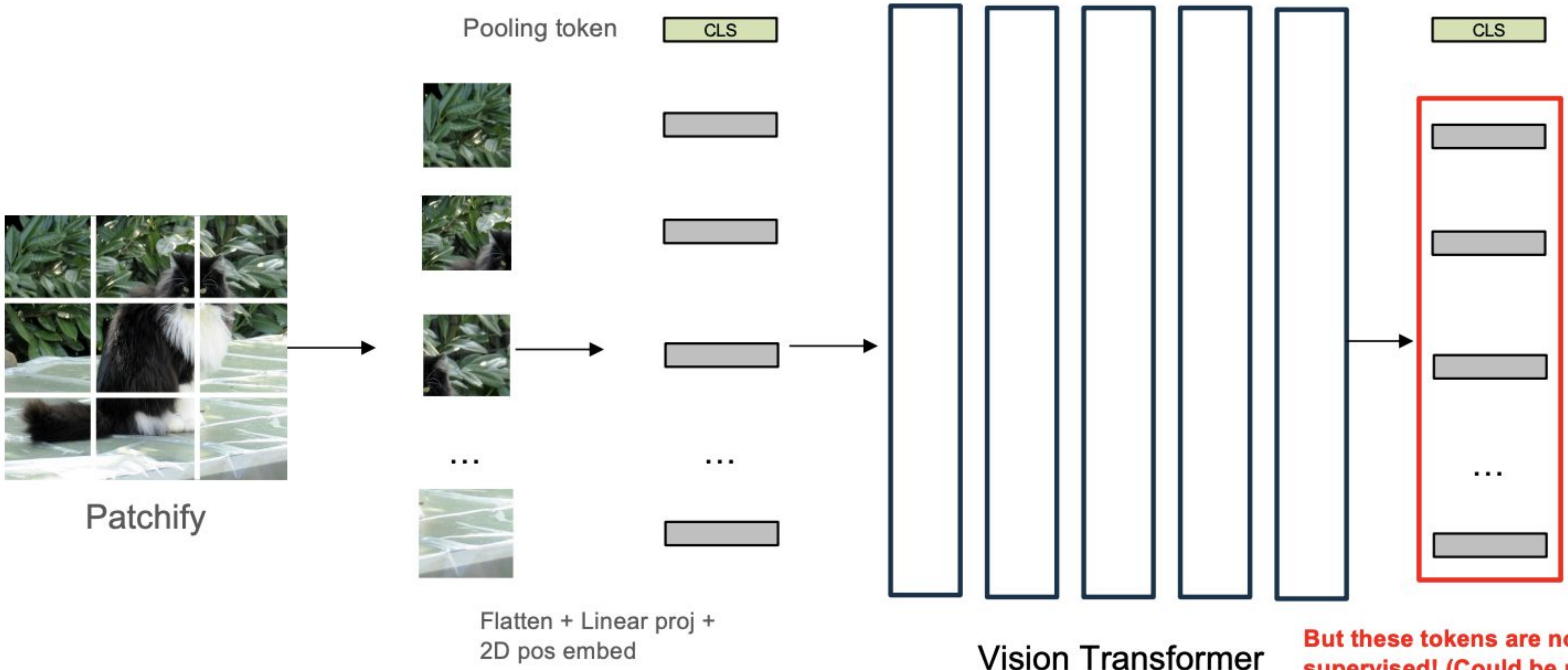
Training: Contrastive Learning



CLIP: Contrastive Learning

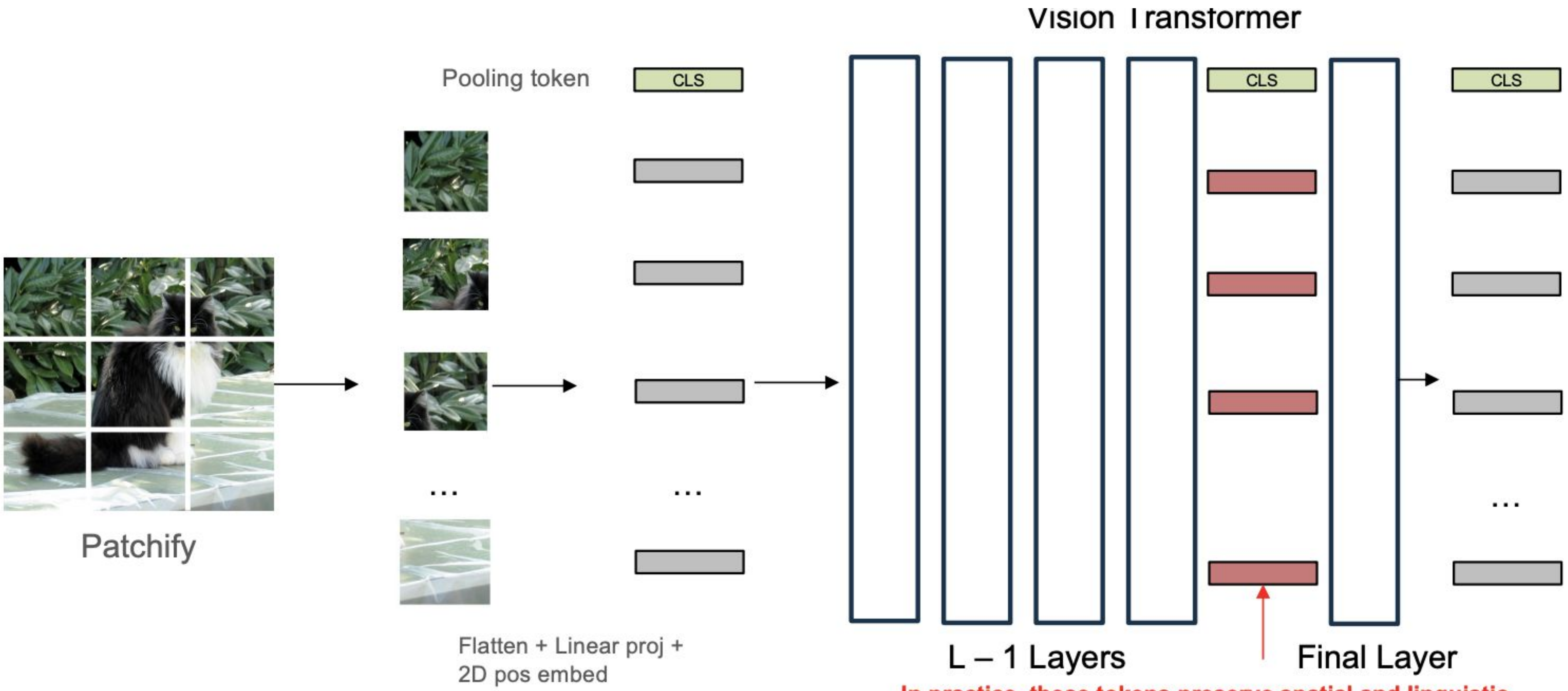


What Feature from CLIP?



But these tokens are not supervised! (Could be random and loss will not change)

What Feature from CLIP?



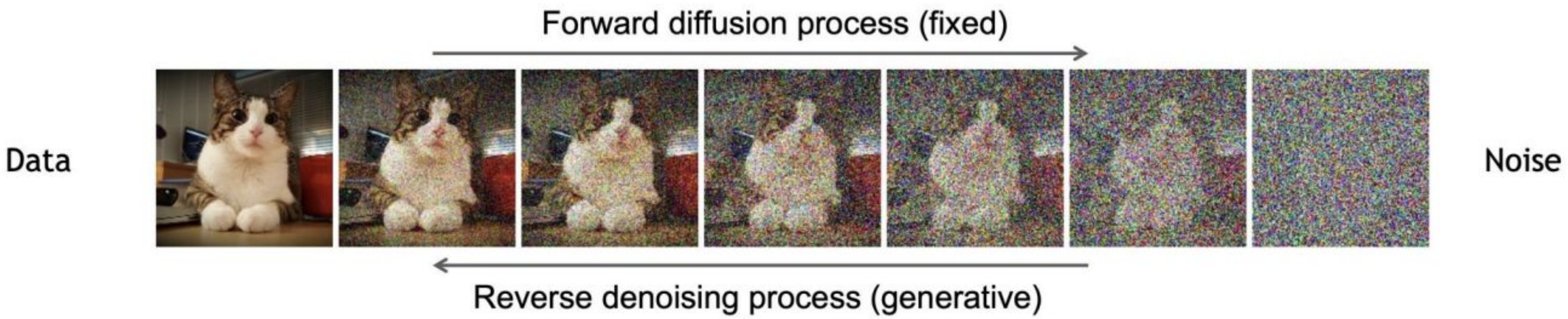
In practice, these tokens preserve spatial and linguistic information best for LLMs. Can drop CLS for slight gains.

Text-to-Image/Video Generation

Video samples generated.

Diffusion Process

- Diffusion models have two processes
- **Forward diffusion process** gradually adds noise to input
- **Reverse denoising process** learns to generate data by denoising



Conditional Diffusion Models

- Un-conditional

$$p\left(\boldsymbol{x}_{0:T}\right)=p\left(\boldsymbol{x}_T\right)\prod_{t=1}^{T}p_{\boldsymbol{\theta}}\left(\boldsymbol{x}_{t-1}\mid\boldsymbol{x}_t\right)$$

- Conditional

$$p\left(\boldsymbol{x}_{0:T}\mid y\right)=p\left(\boldsymbol{x}_T\right)\prod_{t=1}^{T}p_{\boldsymbol{\theta}}\left(\boldsymbol{x}_{t-1}\mid\boldsymbol{x}_t,y\right)$$

More controllable!

Conditional Score Matching

- Score matching with conditional information

$$
\begin{aligned}
\nabla \log p\left(\boldsymbol{x}_t \mid y\right) &= \nabla \log \left(\frac{p\left(\boldsymbol{x}_t\right) p\left(y \mid \boldsymbol{x}_t\right)}{p(y)}\right) \\
&= \nabla \log p\left(\boldsymbol{x}_t\right)+\nabla \log p\left(y \mid \boldsymbol{x}_t\right)-\nabla \log p(y) \\
&= \underbrace{\nabla \log p\left(\boldsymbol{x}_t\right)}_{\text{unconditional score}}+\underbrace{\nabla \log p\left(y \mid \boldsymbol{x}_t\right)}_{\text{adversarial gradient}}
\end{aligned}
$$

Classifier Guidance

- Use a discriminative classifier for $\nabla \log p\left(y \mid \boldsymbol{x}_t\right)$

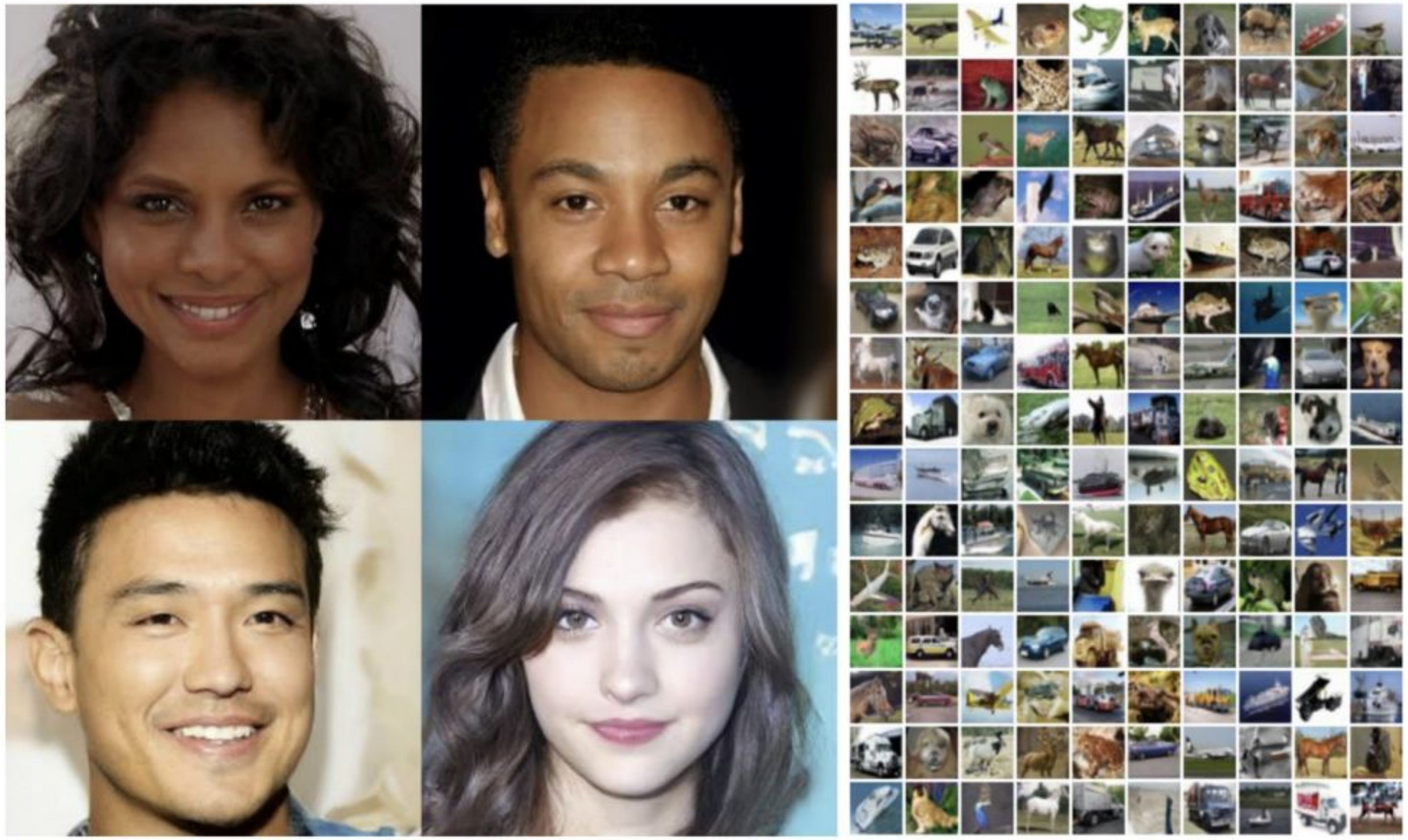
$$\nabla \log p\left(\boldsymbol{x}_t \mid y\right)=\nabla \log p\left(\boldsymbol{x}_t\right)+\gamma \nabla \log p\left(y \mid \boldsymbol{x}_t\right)$$

- γ controls the strength of the condition

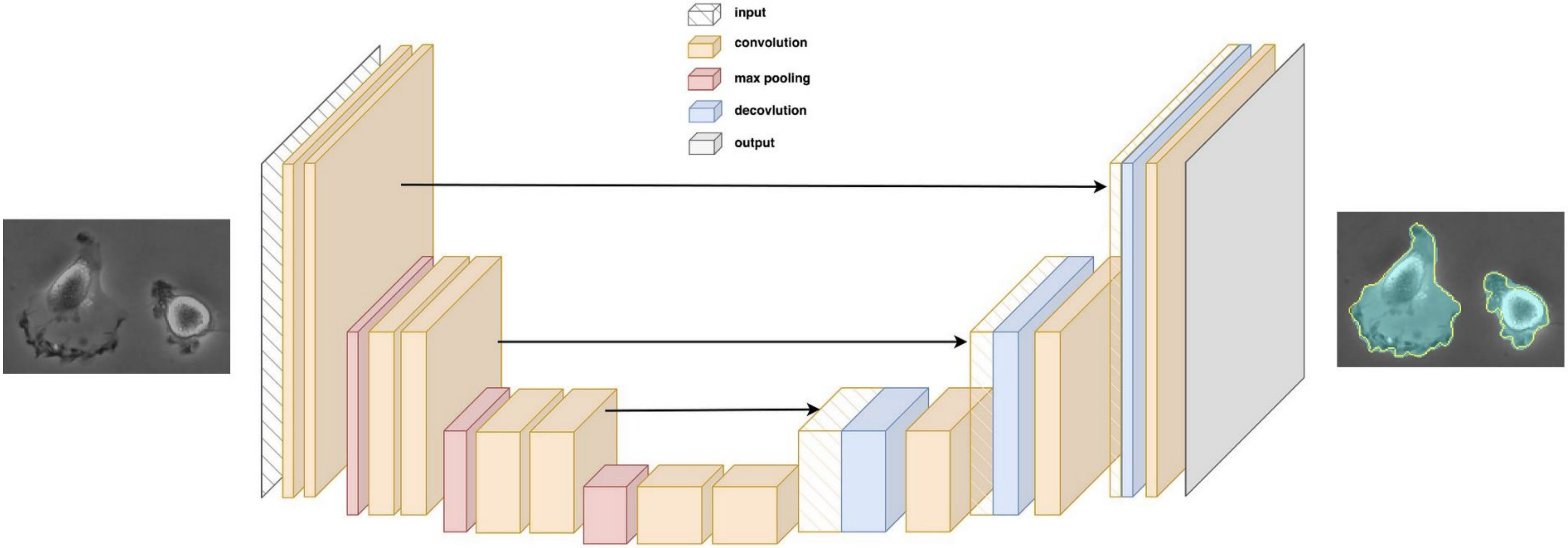
- Limitations:
  - Need a separate classifier
  - Conditioning depends on the performance of classifier

Denoising Diffusion Probabilistic Models (DDPM)

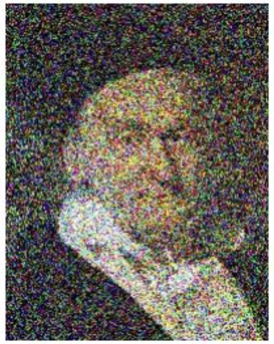
- Training diffusion models on raw images with a U-Net model



The U-Net

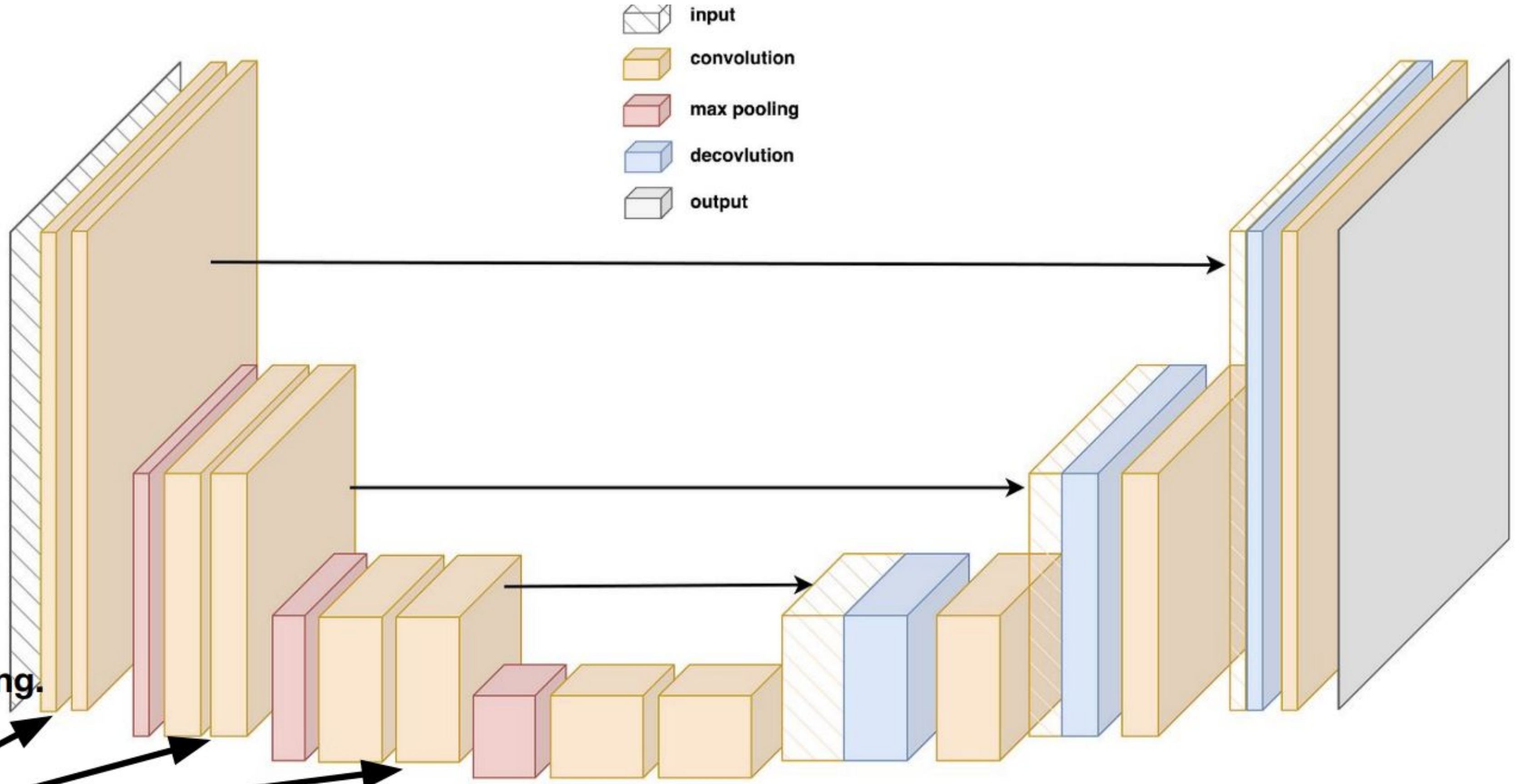


The U-Net



$\sigma = 0.75$

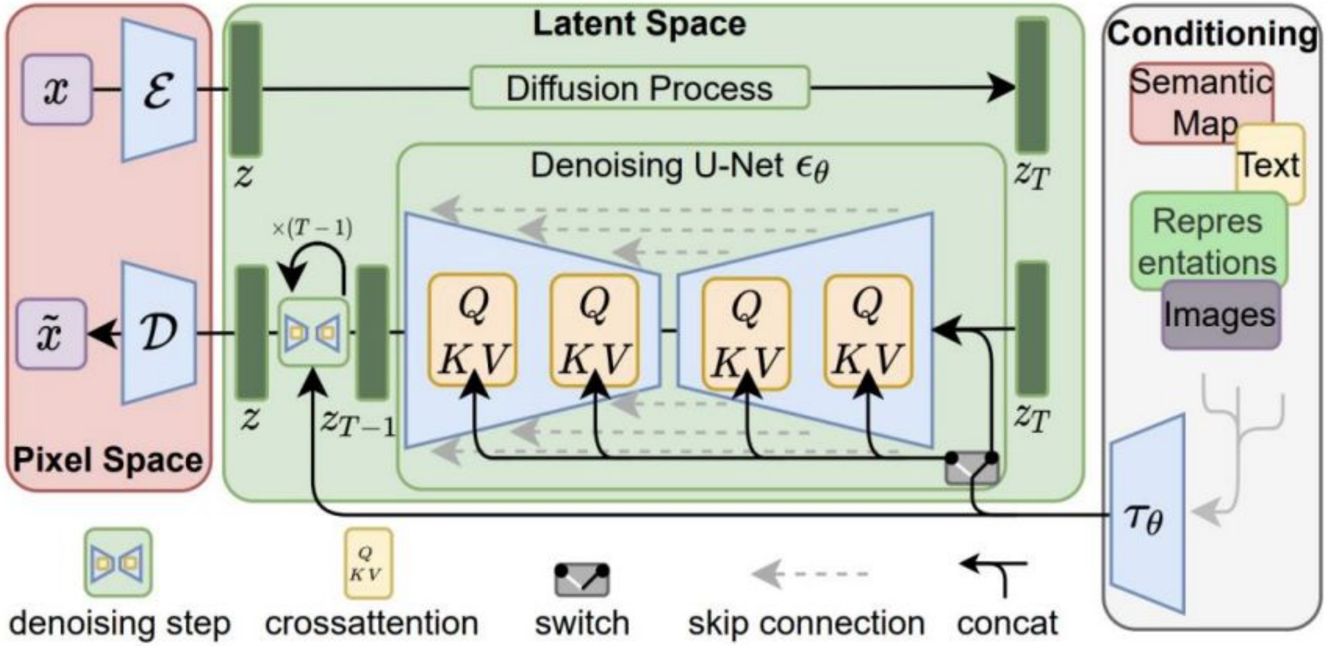
Sinusoidal embedding.



c="An oil painting of a old white man with a white shirt, black hat, and gray side burns."

Latent Diffusion Models (LDMs)

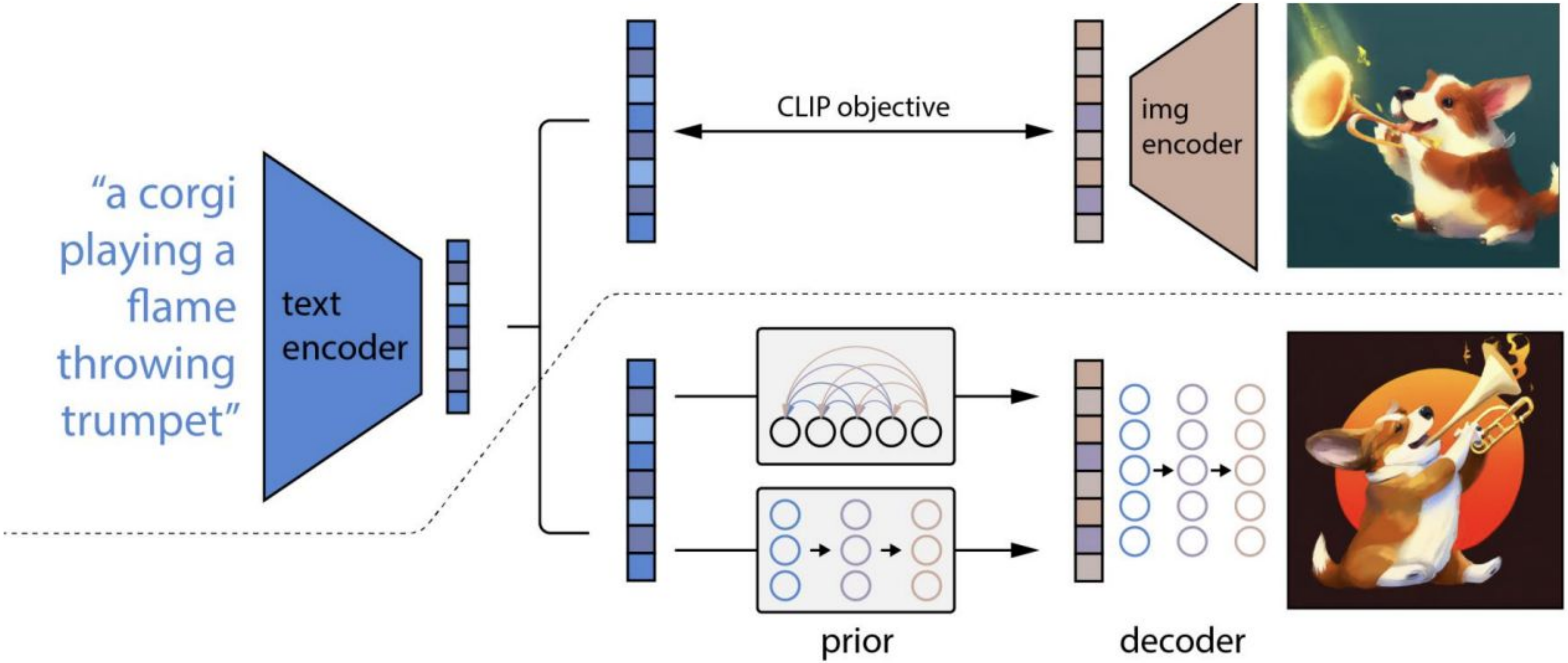
- Learn diffusion on VAE's latent
  - Yet another VAE! Except pre-trained.



Stable Diffusion

- Large-scale text-conditional LDMs
  - With VAEs trained also on larger datasets

DALL-E/DALL-E-2



DALL-E-3

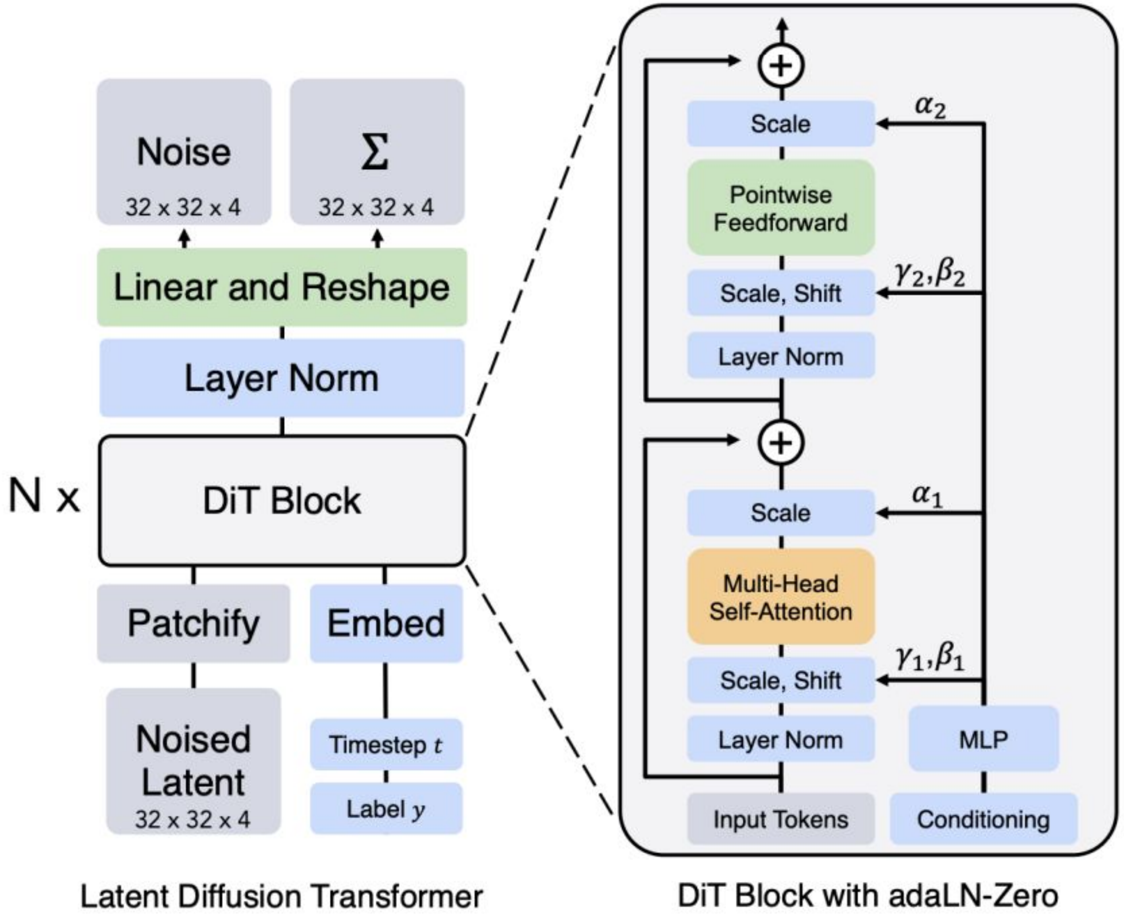
DALL·E 3 makes notable improvements over DALL·E 2, even when given the same prompt.

DALL·E 2 · An expressive oil painting of a basketball player dunking, depicted as an explosion of a nebula.

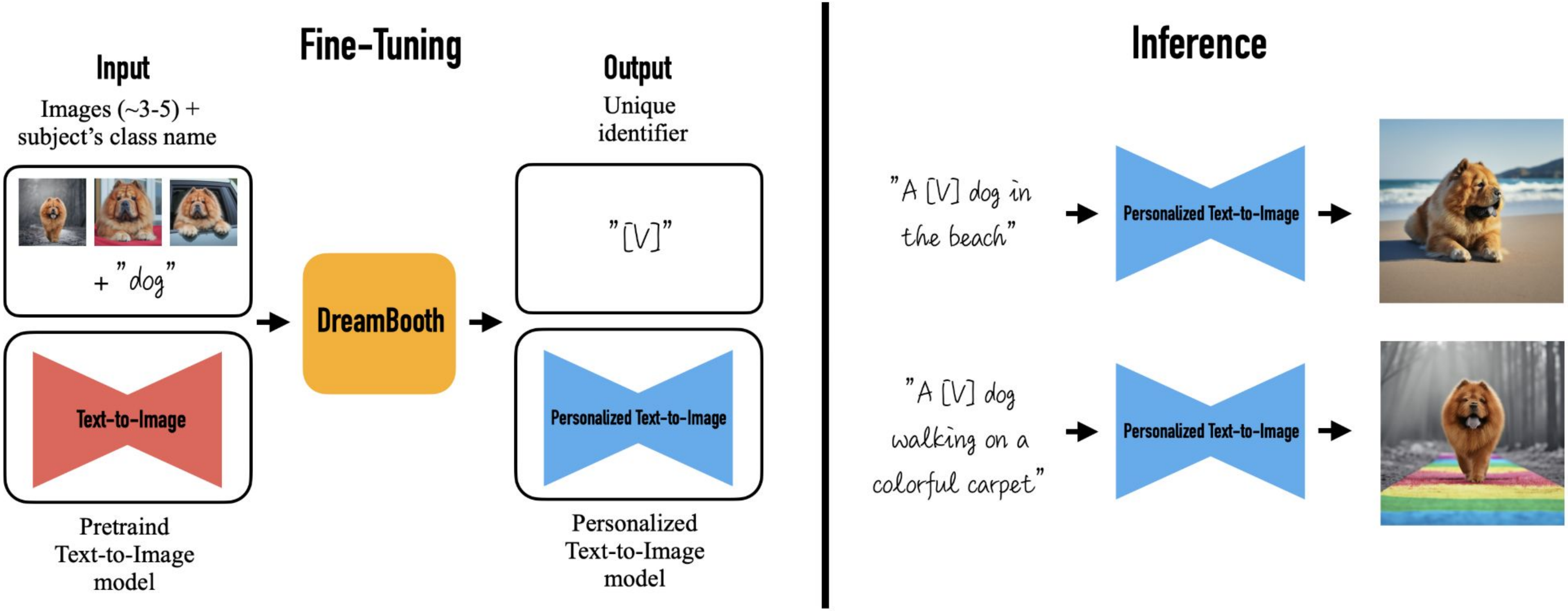
DALL·E 3 · An expressive oil painting of a basketball player dunking, depicted as an explosion of a nebula.

Diffusion Transformer

- A transformer architecture for diffusion models

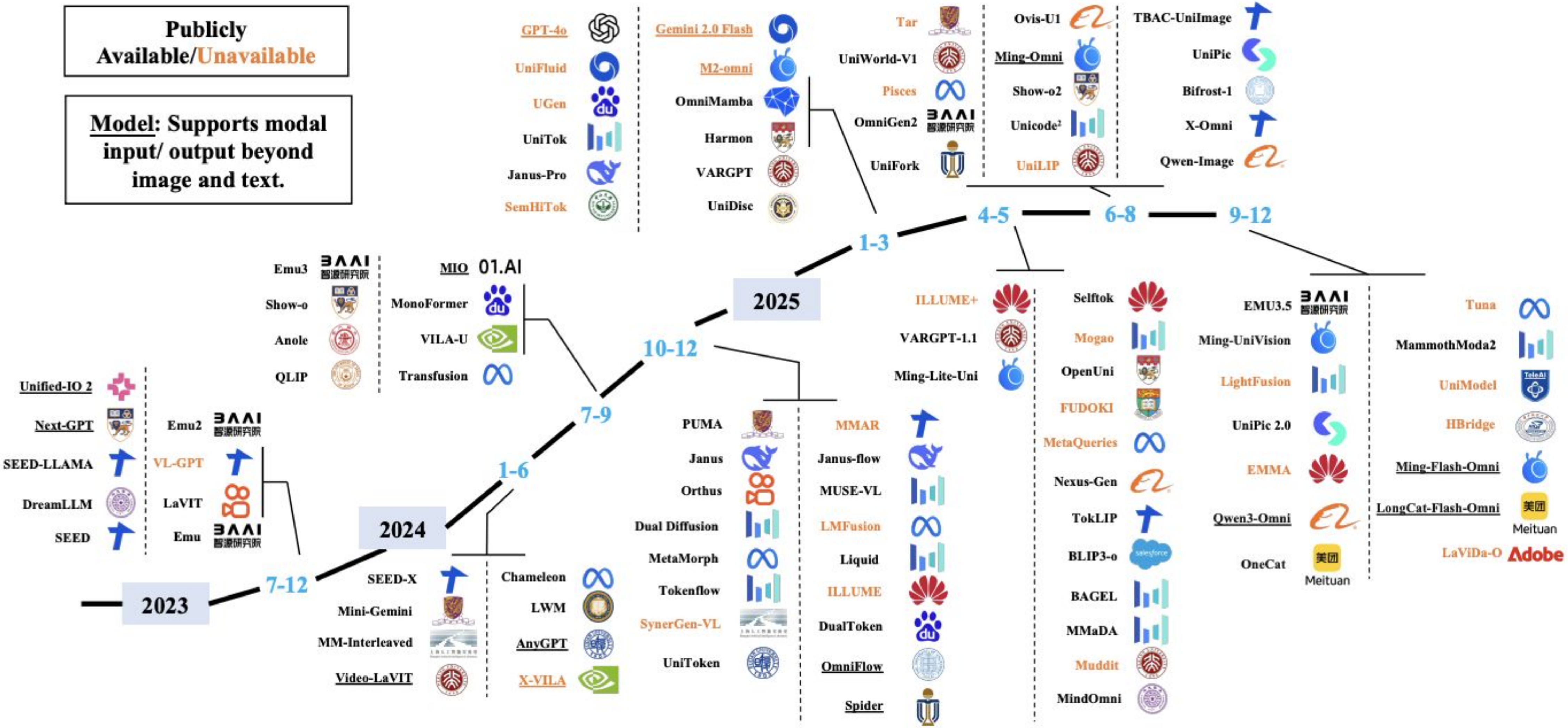


DreamBooth - Subject-Driven Generation

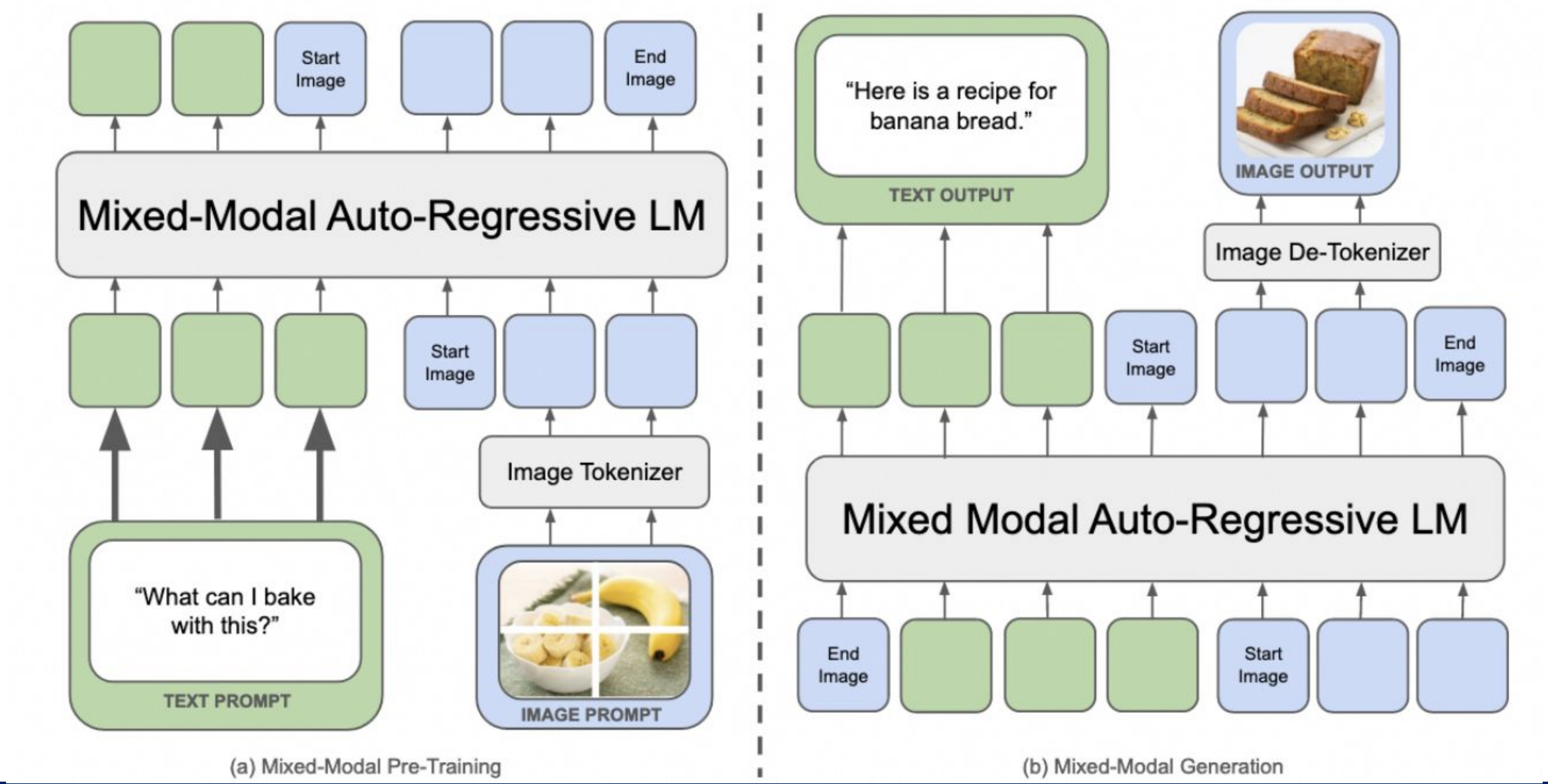


DreamBooth - Subject-Driven Generation

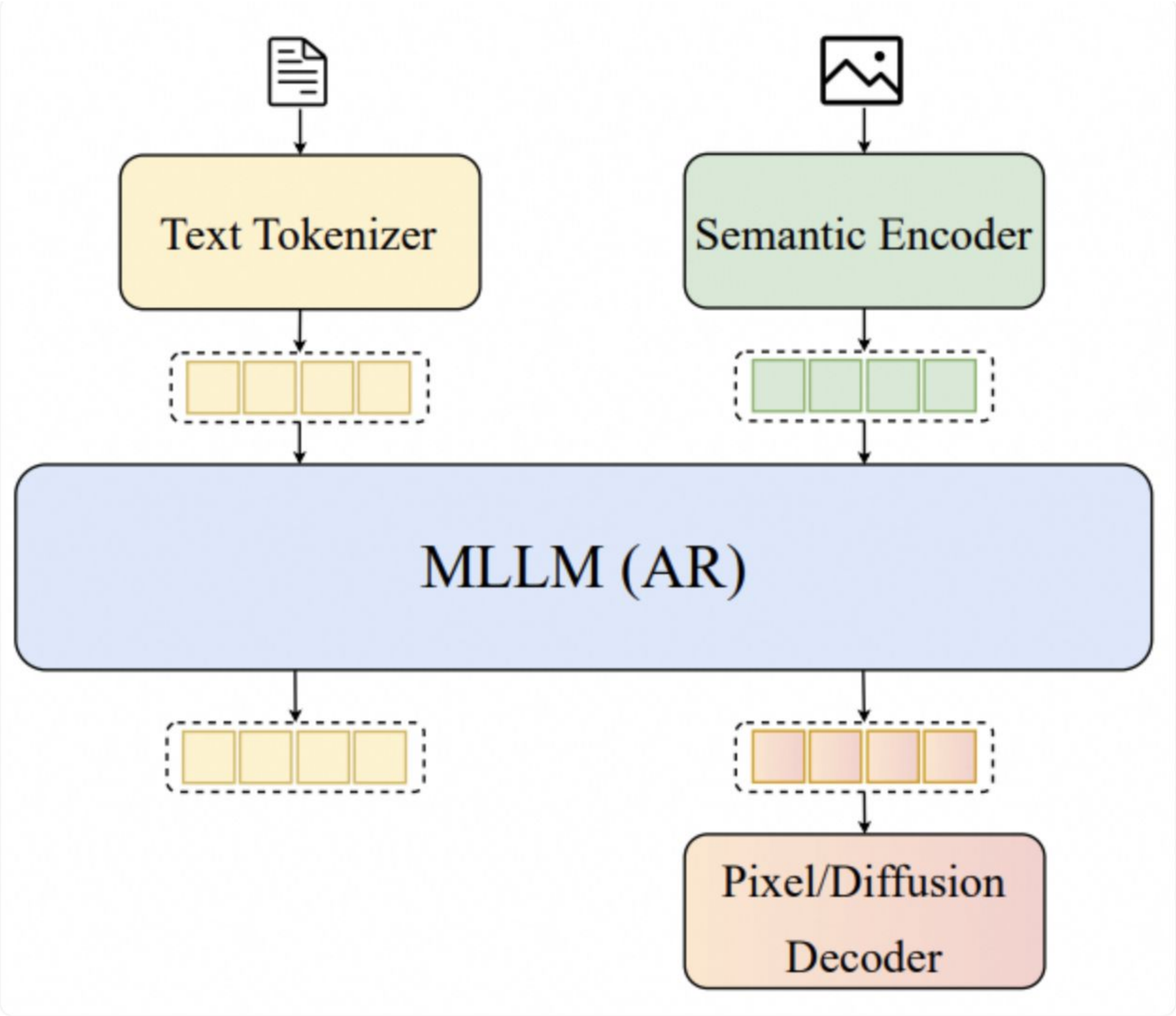
An Unified Modeling - Understanding and Generation



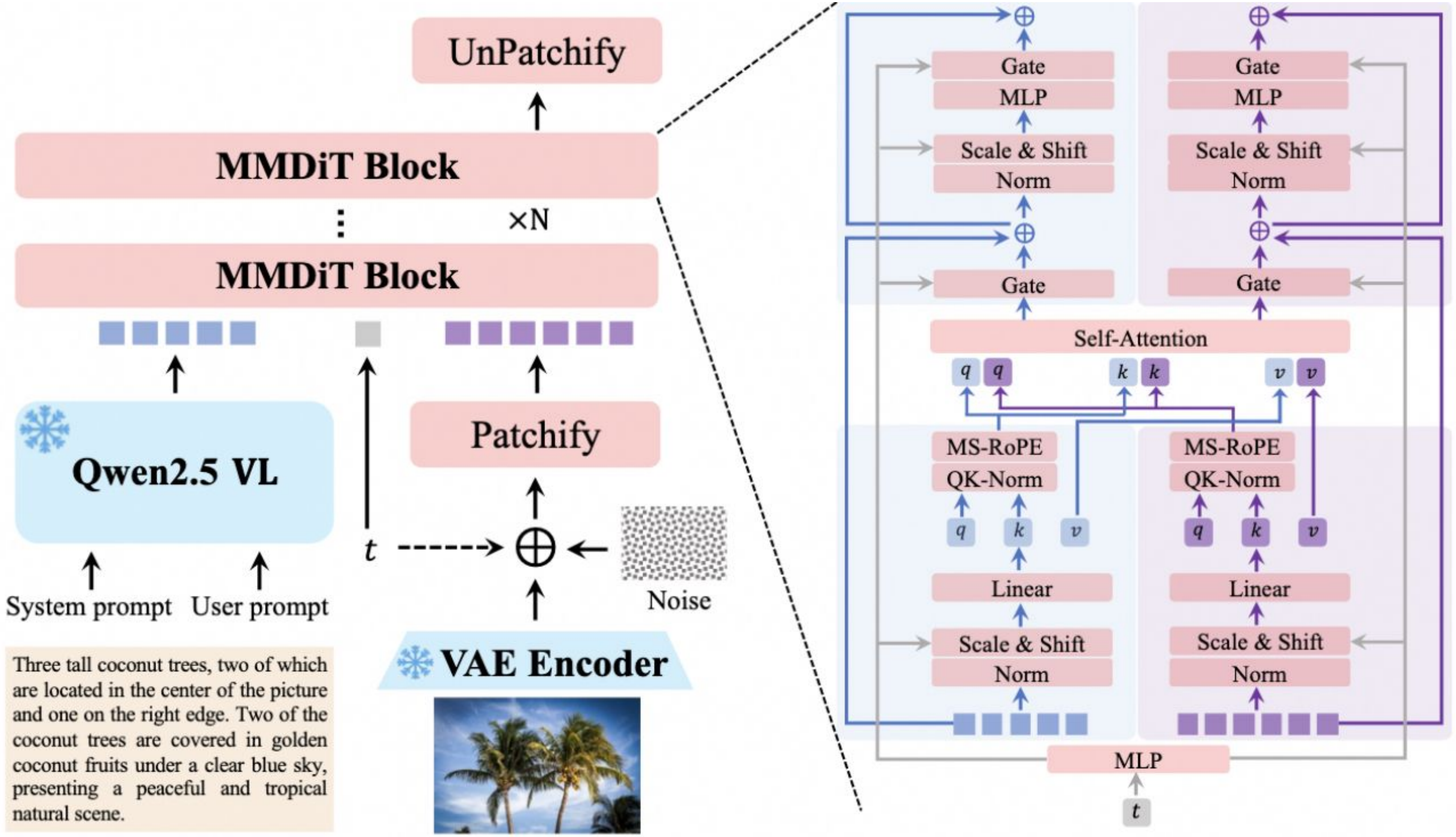
Autoregressive Route Unified Models - Janus-Pro, Emu



AR+Diffusion Serial Structure



Methods Directly Training Diffusion Models - Qwen-Image



MLLM+AR Pixel Encoding - Emu 3.5

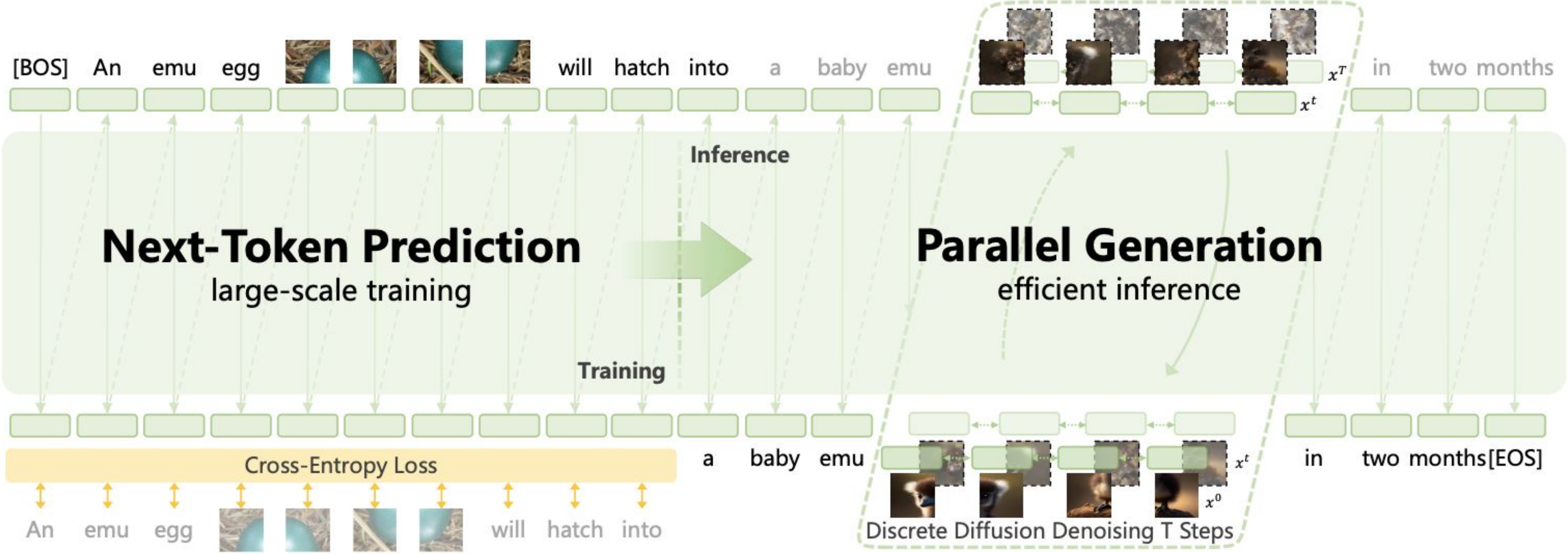
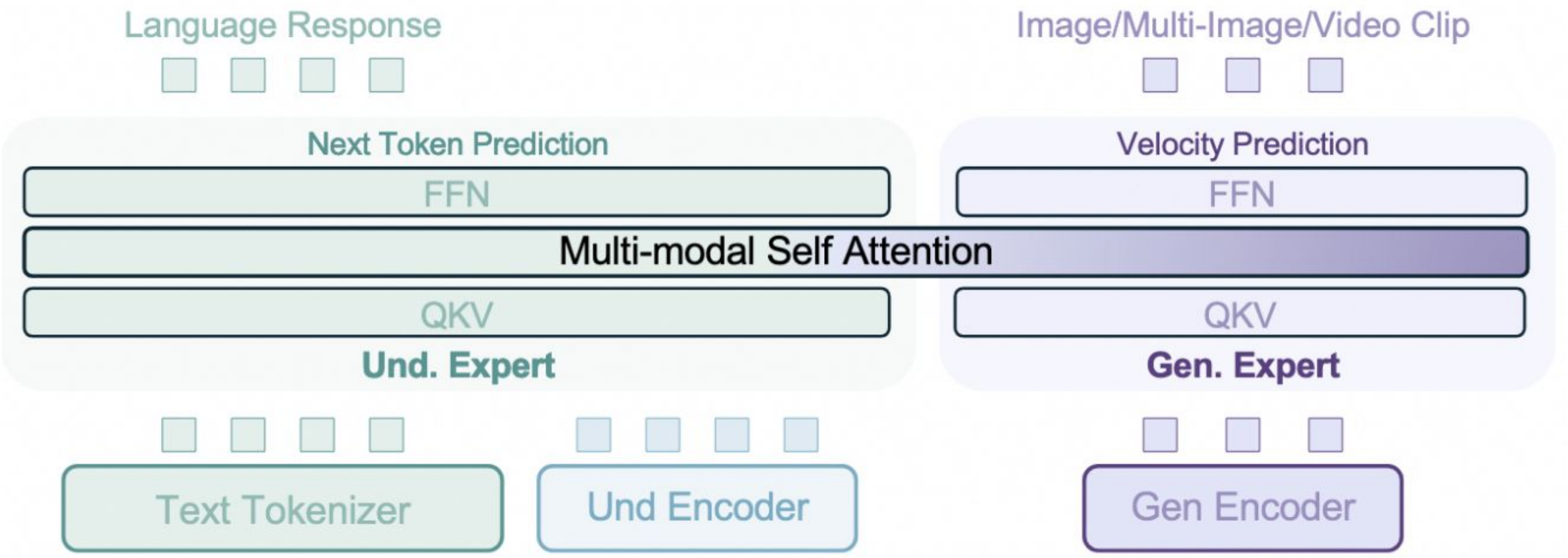
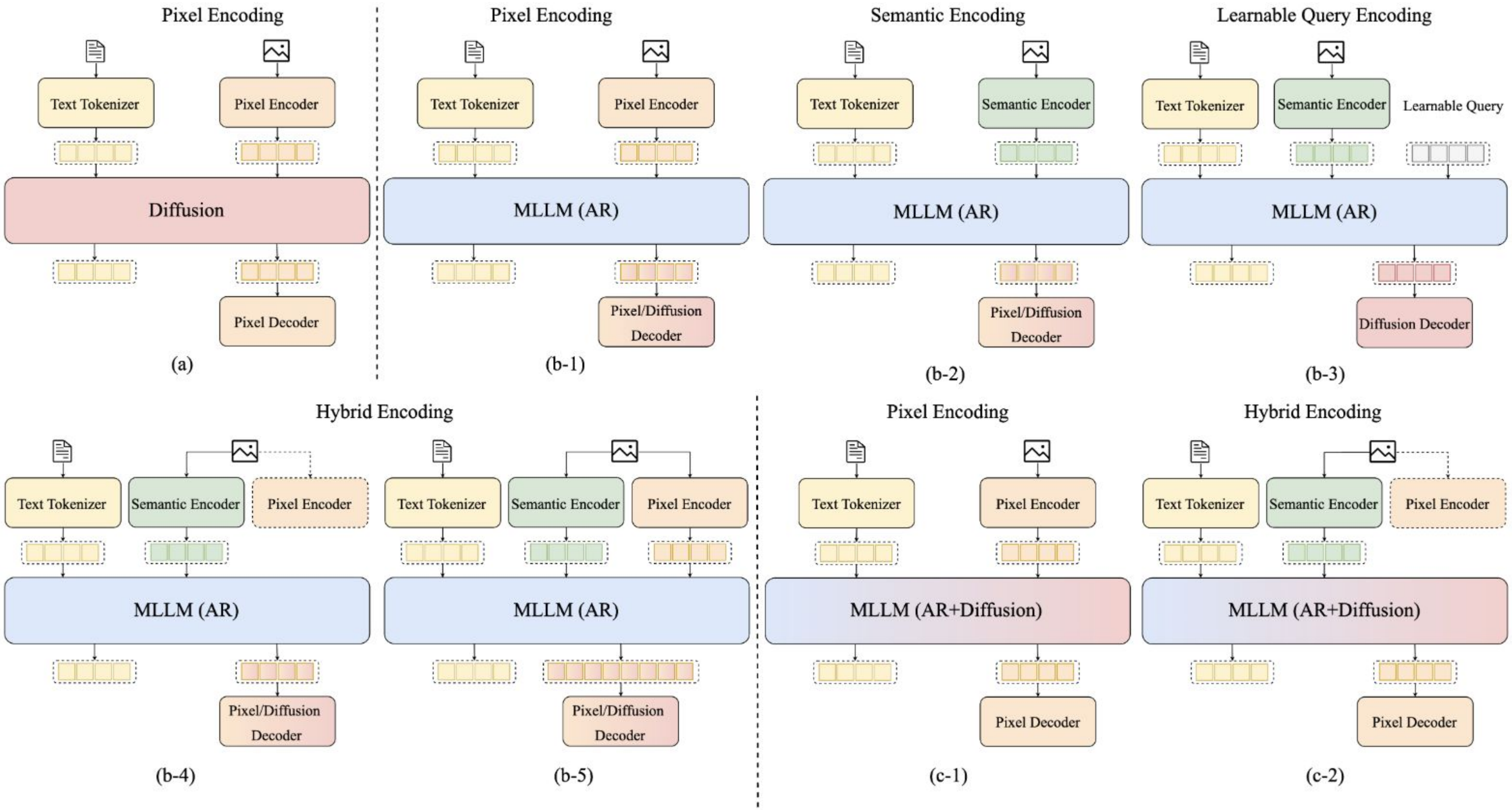


Figure 3: Overview of the Emu3.5 architecture. The model is trained end-to-end at scale with a unified next-token prediction objective. During inference, single-token prediction is accelerated via discrete diffusion adaptation, enabling bidirectional parallel generation per image.

Bagel Mixed Image-Text Training



Text-and-Image Unified Models



Resource - MLLMs

🌋 LLaVA: Large Language and Vision Assistant

Visual instruction tuning towards large language and vision models with GPT-4 level capabilities.

[📢 LLaVA-NeXT Blog] [Project Page] [Demo] [Data] [Model Zoo]

🤝Community Contributions: [llama.cpp] [Colab] [🤗Space] [Replicate] [AutoGen] [BakLLaVA]

Improved Baselines with Visual Instruction Tuning [Paper] [HF]
Haotian Liu, Chunyuan Li, Yuheng Li, Yong Jae Lee

Visual Instruction Tuning (NeurIPS 2023, **Oral**) [Paper] [HF]
Haotian Liu*, Chunyuan Li*, Qingyang Wu, Yong Jae Lee (*Equal Contribution)

Release

- [2024/05/10] 🔥 **LLaVA-NeXT** (Stronger) models are released, stronger LMM with support of LLama-3 (8B) and Qwen-1.5 (72B/110B). [Blog] [Checkpoints] [Demo] [Code]
- [2024/05/10] 🔥 **LLaVA-NeXT** (Video) is released. The image-only-trained LLaVA-NeXT model is surprisingly strong on video tasks with zero-shot modality transfer. DPO training with AI feedback on videos can yield significant improvement. [Blog] [Checkpoints] [Code]
- [03/10] Releasing **LMMs-Eval**, a highly efficient evaluation pipeline we used when developing LLaVA-NeXT. It supports the evaluation of LMMs on dozens of public datasets and allows new dataset onboarding, making the dev of new LMMs much faster. [Blog] [Codebase]

Resource - Text-to-image

Stable Diffusion

Stable Diffusion was made possible thanks to a collaboration with Stability AI and Runway and builds upon our previous work:

High-Resolution Image Synthesis with Latent Diffusion Models
Robin Rombach*, Andreas Blattmann*, Dominik Lorenz, Patrick Esser, Björn Ommer
CVPR '22 Oral | *GitHub* | *arXiv* | *Project page*

Stable Diffusion is a latent text-to-image diffusion model. Thanks to a generous compute donation from Stability AI and support from LAION, we were able to train a Latent Diffusion Model on 512x512 images from a subset of the LAION-5B database. Similar to Google's Imagen, this model uses a frozen CLIP ViT-L/14 text encoder to condition the model on text prompts. With its 860M UNet and 123M text encoder, the model is relatively lightweight and runs on a GPU with at least 10GB VRAM. See this section below and the model card.

Requirements

A suitable conda environment named `ldm` can be created and activated with:

Resource - Text-to-video

OPEN SORA

Stars 29k Gallery View

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Open-Sora: Democratizing Efficient Video Production for All

We design and implement **Open-Sora**, an initiative dedicated to **efficiently** producing high-quality video. We hope to make the model, tools and all details accessible to all. By embracing **open-source** principles, Open-Sora not only democratizes access to advanced video generation techniques, but also offers a streamlined and user-friendly platform that simplifies the complexities of video generation. With Open-Sora, our goal is to foster innovation, creativity, and inclusivity within the field of content creation.

🎬 For a professional AI video-generation product, try Video Ocean — powered by a superior model.

Resource - Text-to-image/Editing

naykun update 6b5e1f5 · 2 weeks ago 86 Commits

assets	update wechat qrcode	5 months ago
src/examples	qwen image 2512 release	2 months ago
LICENSE	Update Sample Code, Readme, and License.	7 months ago
Qwen-Image-Edit-2509.md	Release Qwen-Image-Edit-2509	5 months ago
Qwen-Image-Edit.md	Release Qwen-Image-Edit-2509	5 months ago
Qwen-Image.md	Release Qwen-Image-Edit-2509	5 months ago
README.md	update	2 weeks ago

README Apache-2.0 license

Qwen-Image

💜 Qwen Chat | 🤗 HuggingFace(T2I) | 🤗 HuggingFace(Edit) | 🤖 ModelScope-T2I | 🤖 ModelScope-Edit |
📑 Tech Report | 📑 Blog(T2I) | 📑 Blog(Edit)
🖥️ T2I Demo | 🖥️ Edit Demo | 💬 WeChat (微信) | 🫨 Discord

Resource - Unified Model

BAGEL

BAGEL Website | BAGEL Paper | BAGEL Model | BAGEL Demo | BAGEL Space | BAGEL Discord | BAGEL Email

Unified Model for Multimodal Understanding and Generation

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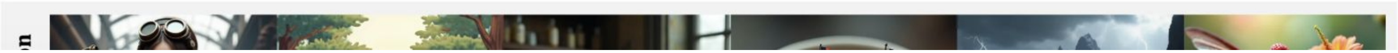
We present **BAGEL**, an open-source multimodal foundation model with 7B active parameters (14B total) trained on large-scale interleaved multimodal data. BAGEL outperforms the current top-tier open-source VLMs like Qwen2.5-VL and InternVL-2.5 on standard multimodal understanding leaderboards, and delivers text-to-image quality that is competitive with strong specialist generators such as SD3. Moreover, BAGEL demonstrates superior qualitative results in classical image-editing scenarios than the leading open-source models. More importantly, it extends to free-form visual manipulation, multiview synthesis, and world navigation, capabilities that constitute "world-modeling" tasks beyond the scope of previous image-editing models. The figure below showcases BAGEL's qualitative performance.



Resource - Unified Model

🚀 Janus-Series: Unified Multimodal Understanding and Generation Models

DeepSeek Homepage | Hugging Face DeepSeek AI

Code License MIT | Model License Model Agreement

📥 Model Download | ⚡ Quick Start | 📜 License | 📖 Citation

🤗 Online Demo (Janus-Pro-7B, Janus, JanusFlow)

News

2025.01.27: Janus-Pro is released, an advanced version of Janus, improving both multimodal understanding and visual generation significantly. See paper

2024.11.13: JanusFlow is released, a new unified model with rectified flow for image generation. See paper, demo and usage.

2024.10.23: Evaluation code for reproducing the multimodal understanding results from the paper has been added to VLMEvalKit. Please refer to this link.

2024.10.20: (1) Fix a bug in tokenizer_config.json. The previous version caused classifier-free guidance to not function properly, resulting in relatively poor visual generation quality. (2) Release Gradio demo (online demo and local).

1. Introduction

Janus-Pro: Unified Multimodal Understanding and Generation with Data and Model Scaling

Thanks

Q&A

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